Diss cleaning

SCL

2026-08-12

```
library(pwr)
library(tinytex)
```

```
## Warning: package 'tinytex' was built under R version 4.5.3
```

```
library(sjstats)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tibble)
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.3
```

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ forcats   1.0.1     ✓ readr     2.1.6
## ✓ ggplot2   4.0.2     ✓ stringr   1.5.2
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.2
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(here)
```

```
## here() starts at C:/Users/slewi/OneDrive/Documents/GitHub/Perception-of-researcher-roles
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
##
## The following object is masked from 'package:sjstats':
##
##     phi
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:psych':
##
##     logit
##
## The following object is masked from 'package:purrr':
##
##     some
##
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(effectsize)
```

```
##
## Attaching package: 'effectsize'
##
## The following object is masked from 'package:psych':
##
##     phi
##
## The following objects are masked from 'package:sjstats':
##
##     cohens_f, cramers_v, phi
```

```
library(broom)
```

```
##
## Attaching package: 'broom'
##
## The following object is masked from 'package:sjstats':
##
##     bootstrap
```

```
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
library(lattice)
library(nlme)
```

```
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```
library(lubridate)
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
##
## Attaching package: 'lme4'
##
## The following object is masked from 'package:nlme':
##
##     lmList
```

```
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.5.3
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```
style_table = function(kbl, footnote_text = NULL, header_above = NULL) {
  if (knitr::is_html_output()) {
    kbl = kableExtra::kable_styling(kbl, full_width = FALSE,
                                    bootstrap_options = "condensed")
    if (!is.null(header_above)) {
      kbl = kableExtra::add_header_above(kbl, header_above)
    }
    if (!is.null(footnote_text)) {
      kbl = kableExtra::footnote(kbl, general = footnote_text,
                                 general_title = "Note.",
                                 footnote_as_chunk = TRUE)
    }
    kbl
  } else {
    if (!is.null(footnote_text)) {
      cat(paste0("\n\n*Note.* ", footnote_text, "\n\n"))
    }
    kbl
  }
}
```

###LOAD DATA

```
library(here)
data_path = here::here("Diss_full.csv")
diss.full =  read.csv(data_path)
```

####LET'S START CLEANING

```
#make all the columns lowercase to help with coding
names(diss.full) = tolower(names(diss.full))
names(diss.full) =  iconv(names(diss.full), from = "", to = "UTF-8", sub = "")
names(diss.full) =  gsub("consent.*", "consent", names(diss.full))
```

Okay data is loaded. Now, we will clean it by the guidelines of the pre-reg.

Participants will be excluded if they: 1) do not consent or 2) complete the study in less than seven minutes or 3) exhibit no variability in their responses (e.g., selecting all 5s or all 3s) across all Integrity and likeability items, and 4) fail to complete the Integrity and likeability ratings for at least one vignettes.

An attention check will be used to assess whether participants paid attention during the experiment by requiring participants to select a specific response to the question (i.e., "Select the option "somewhat disagree" if you are paying attention."). We will do an exploratory analysis to determine test if excluding those who fail the attention impacts the data.

```
# Initial data count
initial_count =  nrow(diss.full)
print(initial_count)
```

```
## [1] 675
```

```
# Clean recordeddate
diss.full$recordeddate =  str_squish(as.character(diss.full$recordeddate))
diss.full$created_date = mdy_hm(diss.full$recordeddate)

# Filters by cutoff date
cutoff_date =  as.Date("2026-01-27")
filtered_by_date = diss.full %>%
  filter(!is.na(created_date) & created_date >= cutoff_date)

count_by_date =  nrow(filtered_by_date)
print(count_by_date)
```

```
## [1] 675
```

```r
# Filter by consent
filtered_by_consent = filtered_by_date %>%
  filter(consent == "Yes, I consent")

count_by_consent =  nrow(filtered_by_consent)

# Filter by duration (exclude under 420 seconds)
filtered_by_duration = filtered_by_consent %>%
  filter(!is.na(duration..in.seconds.) & duration..in.seconds. >= 420)

count_by_duration = nrow(filtered_by_duration)

# Exclude participants under 18
filtered_by_age = filtered_by_duration %>%
  filter(is.na(age) | age >= 18)

count_by_age = nrow(filtered_by_age)


# Exclude no-variability: participants who gave identical responses across
# all Integrity and likeability items — uses apply() to avoid rowwise() issues
Integrity_like_cols = c("khon","ksin","keth","klike","kapp",
                    "shon","ssin","seth","slike","sapp",
                    "hhon","hsin","heth","hlike","happ",
                    "fhon","fsin","feth","flike","fapp",
                    "xhon","xsin","xeth","xlike","xapp",
                    "yhon","ysin","yeth","ylike","yapp")

tl_matrix = filtered_by_age %>%
  select(all_of(Integrity_like_cols)) %>%
  mutate(across(everything(), as.numeric))
```

```
## Warning: There were 30 warnings in `mutate()`.
## The first warning was:
## ℹ In argument: `across(everything(), as.numeric)`.
## Caused by warning:
## ! NAs introduced by coercion
## ℹ Run `dplyr::last_dplyr_warnings()` to see the 29 remaining warnings.
```

```
has_variability = apply(tl_matrix, 1, function(row) n_distinct(na.omit(row)) > 1)
filtered_by_variability = filtered_by_age[has_variability, ]
count_by_variability = nrow(filtered_by_variability)

# Exclude attention check failures (attention1 == 2 is correct response)
filtered_by_attention = filtered_by_variability %>%
  filter(attention1 == 2)
count_by_attention = nrow(filtered_by_attention)
# Final count
final_count = nrow(filtered_by_age)

# Summary table
count_summary = data.frame(
  Step = c(
    "Initial",
    "After Date Filter",
    "After Consent Filter",
    "After Duration Filter",
    "After Age Filter",
    "Final"
  ),
  Count = c(
    initial_count,
    count_by_date,
    count_by_consent,
    count_by_duration,
    count_by_age,
    final_count
  )
)

print(count_summary)
```

```
##                    Step Count
## 1               Initial   675
## 2     After Date Filter   675
## 3  After Consent Filter   675
## 4 After Duration Filter   455
## 5      After Age Filter   455
## 6                 Final   455
```

```
# Print counts
cat("Total rows in diss.full:", initial_count, "\n")
```

```
## Total rows in diss.full: 675
```

```
cat("Rows after filtering by date:", count_by_date, "\n")
```

```
## Rows after filtering by date: 675
```

```
cat("Rows after filtering by consent:", count_by_consent, "\n")
```

```
## Rows after filtering by consent: 675
```

```
cat("Rows after filtering by duration:", count_by_duration, "\n")
```

```
## Rows after filtering by duration: 455
```

```
cat("Rows after excluding participants under 18:", count_by_age, "\n")
```

```
## Rows after excluding participants under 18: 455
```

```
cat("Final rows:", final_count, "\n")
```

```
## Final rows: 455
```

okay now we filter by ege

```
attr_dat = filtered_by_age %>%
  select(
    responseid,
    minsmart, minbetter, mincreative, minrather, minshould, minregard, mintyp, minkeep, minget,
    modsmart, modbett, modcreat, modrath, modshou, modreg, modtyp, modkeep, modget,
    majsmart, majbet, majcreat, majrath, majshould, majreg, majtyp, majkeep, majget
  )
```

##DEMO

```
# AGE
filtered_by_age$age <- as.numeric(as.character(filtered_by_age$age))

unique_age <- sort(unique(filtered_by_age$age))
print(unique_age)
```

```
## [1] 18 19 20 21 22 23 24 26 36
```

```
mean_age <- mean(filtered_by_age$age, na.rm = TRUE)
print(mean_age)
```

```
## [1] 19.28182
```

```
sd_age <- sd(filtered_by_age$age, na.rm = TRUE)
print(sd_age)
```

```
## [1] 1.423089
```

```
age_range <- range(filtered_by_age$age, na.rm = TRUE)
print(age_range)
```

```
## [1] 18 36
```

```
# CONSENT
consent_table <- table(
  filtered_by_age$consent,
  useNA = "ifany"
)
print(consent_table)
```

```
##
## Yes, I consent
##            455
```

```
# DEMOGRAPHIC FREQUENCY TABLES
gender_table <- table(
  filtered_by_age$gender.identity,
  useNA = "ifany"
)

ethnicity_table <- table(
  filtered_by_age$ethnicity,
  useNA = "ifany"
)

SO_table <- table(
  filtered_by_age$sex.orientation,
  useNA = "ifany"
)

race_table <- table(
  filtered_by_age$race,
  useNA = "ifany"
)

political_table <- table(
  filtered_by_age$political,
  useNA = "ifany"
)

religion_table <- table(
  filtered_by_age$religion,
  useNA = "ifany"
)


# PRINT COUNTS
print(gender_table)
```

```
##
##                                          Man Non-binary / third gender
##                        15                 82                         2
##         Prefer not to say              Woman
##                         2                354
```

```
print(ethnicity_table)
```

```
##
##            Don’t Know         No        Yes
##         15          3        397         40
```

```
print(SO_table)
```

```
##
##                                  Asexual             Bisexual
##                    16                  3                   18
## Heterosexual/straight      Homosexual (gay)             Lesbian
##                   402                  3                    5
##                 Other     Prefer not to say                Queer
##                     1                  5                    2
```

```
print(race_table)
```

```
##
##
##                                             15
##                             Asian or Asian-American
##                                              7
##                           Black or African-American
##                                             87
## Native American, American Indian, or Alaska Native
##                                              2
##                                          Other
##                                             14
##                                          White
##                                            330
```

```
print(political_table)
```

```
##
##                                  Centrist          Conservative
##                     26                    90                   111
##     Lean Conservative           Lean Liberal               Liberal
##                     66                    37                    61
## Strongly Conservative      Strongly Liberal
##                     44                    20
```

```
print(religion_table)
```

```
##
##                                       Catholicism                     Islam
##                        17                        89                         1
##                   Judaism            Not Affiliated                     Other
##                         2                        59                        28
## Protestantism (Christian)
##                       259
```

```
# DEMOGRAPHIC PERCENTAGES
gender_percent <- prop.table(gender_table) * 100
ethnicity_percent <- prop.table(ethnicity_table) * 100
SO_percent <- prop.table(SO_table) * 100
race_percent <- prop.table(race_table) * 100
political_percent <- prop.table(political_table) * 100
religion_percent <- prop.table(religion_table) * 100


# PRINT PERCENTAGES
print(round(gender_percent, 2))
```

```
##
##                                            Man Non-binary / third gender
##                      3.30                    18.02                      0.44
##         Prefer not to say                    Woman
##                      0.44                    77.80
```

```
print(round(ethnicity_percent, 2))
```

```
##
##            Don’t Know         No        Yes
##       3.30       0.66      87.25       8.79
```

```
print(round(SO_percent, 2))
```

```
##
##                                  Asexual              Bisexual
##                     3.52             0.66                  3.96
## Heterosexual/straight      Homosexual (gay)               Lesbian
##                    88.35             0.66                  1.10
##                    Other     Prefer not to say               Queer
##                     0.22             1.10                  0.44
```

```
print(round(race_percent, 2))
```

```
##
##
##                                           3.30
##                             Asian or Asian-American
##                                           1.54
##                          Black or African-American
##                                          19.12
## Native American, American Indian, or Alaska Native
##                                           0.44
##                                          Other
##                                           3.08
##                                          White
##                                          72.53
```

```
print(round(political_percent, 2))
```

```
##
##                                 Centrist          Conservative
##                     5.71            19.78                 24.40
##     Lean Conservative          Lean Liberal               Liberal
##                    14.51             8.13                 13.41
## Strongly Conservative      Strongly Liberal
##                     9.67             4.40
```

```
print(round(religion_percent, 2))
```

```
##
##                                    Catholicism                    Islam
##                      3.74                  19.56                     0.22
##                   Judaism           Not Affiliated                    Other
##                      0.44                  12.97                     6.15
## Protestantism (Christian)
##                     56.92
```

```
# OTHER RESPONSES
unique_rel <- unique(
  na.omit(filtered_by_age$religion.other)
)
print(unique_rel)
```

```
##  [1] ""
##  [2] "Christian / Non-denominational "
##  [3] "Baptist"
##  [4] "unsure"
##  [5] "presbyterian"
##  [6] "Non denominational "
##  [7] "Church of Jesus Christ of Latter Day Saints"
##  [8] "Catholic "
##  [9] "N/A"
## [10] "christianity"
## [11] "Christian"
## [12] "sikh"
## [13] "Baptist "
## [14] "Christian (spiritual)"
## [15] "agnostic"
```

```
unique_race <- unique(
  na.omit(filtered_by_age$race.other)
)
print(unique_race)
```

```
##  [1] ""
##  [2] "mixed"
##  [3] "Mixed- white and black"
##  [4] "mixed race, black and white "
##  [5] "African-American and White"
##  [6] "Biracial; African American and Japanese "
##  [7] "asian american"
##  [8] "White and east asian mix. "
##  [9] "latino "
## [10] "White, and Asian (Korean)"
## [11] "Mexican"
## [12] "Guatemalan/American"
```

```
unique_sexual_orientation <- unique(
  na.omit(filtered_by_age$so.other)
)
print(unique_sexual_orientation)
```

```
## [1] ""
## [2] "No Labeler; prefer not to put a label on my sexuality."
```

```
unique_gender <- unique(
  na.omit(filtered_by_age$gender.other)
)
print(unique_gender)
```

```
## logical(0)
```

```r
# COUNTS + PERCENTAGES IN ONE TABLE
demo_table <- function(x) {
  counts <- table(x, useNA = "ifany")
  percentages <- prop.table(counts) * 100

  data.frame(
    Response = names(counts),
    N = as.vector(counts),
    Percent = round(as.vector(percentages), 2)
  )
}


# CREATE DEMOGRAPHIC TABLES
gender_demo <- demo_table(filtered_by_age$gender.identity)
ethnicity_demo <- demo_table(filtered_by_age$ethnicity)
race_demo <- demo_table(filtered_by_age$race)
SO_demo <- demo_table(filtered_by_age$sex.orientation)
political_demo <- demo_table(filtered_by_age$political)
religion_demo <- demo_table(filtered_by_age$religion)


# PRINT COUNTS + PERCENTAGES
print(gender_demo)
```

```
##                    Response   N Percent
## 1                           15    3.30
## 2                       Man  82   18.02
## 3 Non-binary / third gender   2    0.44
## 4         Prefer not to say   2    0.44
## 5                     Woman 354   77.80
```

```r
print(ethnicity_demo)
```

```
##     Response   N Percent
## 1             15    3.30
## 2 Don’t Know   3    0.66
## 3         No 397   87.25
## 4        Yes  40    8.79
```

```r
print(race_demo)
```

```
##                                          Response   N Percent
## 1                                                  15    3.30
## 2                          Asian or Asian-American   7    1.54
## 3                        Black or African-American  87   19.12
## 4 Native American, American Indian, or Alaska Native   2    0.44
## 5                                           Other  14    3.08
## 6                                           White 330   72.53
```

```r
print(SO_demo)
```

```
##                Response   N Percent
## 1                         16    3.52
## 2               Asexual   3    0.66
## 3              Bisexual  18    3.96
## 4 Heterosexual/straight 402   88.35
## 5      Homosexual (gay)   3    0.66
## 6               Lesbian   5    1.10
## 7                 Other   1    0.22
## 8     Prefer not to say   5    1.10
## 9                 Queer   2    0.44
```

```r
print(political_demo)
```

```
##                Response   N Percent
## 1                        26    5.71
## 2             Centrist  90   19.78
## 3          Conservative 111   24.40
## 4     Lean Conservative  66   14.51
## 5          Lean Liberal  37    8.13
## 6               Liberal  61   13.41
## 7 Strongly Conservative  44    9.67
## 8      Strongly Liberal  20    4.40
```

```
print(religion_demo)
```

```
##                    Response   N Percent
## 1                          17    3.74
## 2               Catholicism  89   19.56
## 3                     Islam   1    0.22
## 4                   Judaism   2    0.44
## 5            Not Affiliated  59   12.97
## 6                     Other  28    6.15
## 7 Protestantism (Christian) 259   56.92
```

###Attribute

```
# Pivot the dataset

attr_long = attr_dat %>%
  pivot_longer(
    cols = -responseid,
    names_to = "raw_item",
    values_to = "choice_raw"
  )


# Find the correct columns to attribute to the variables

attr_long = attr_long %>%
  mutate(
    severity = case_when(
      str_detect(raw_item, "^min") ~ "minor",
      str_detect(raw_item, "^mod") ~ "moderate",
      str_detect(raw_item, "^maj") ~ "major",
      TRUE ~ NA_character_
    ),
    attribute = case_when(
      str_detect(raw_item, "smart$") ~ "smarter",
      str_detect(raw_item, "bett(er)?$|bet$") ~ "better_researcher",
      str_detect(raw_item, "creat(ive)?$") ~ "creative",
      str_detect(raw_item, "rath(er)?$") ~ "rather_be",
      str_detect(raw_item, "shou(ld)?$") ~ "should_be",
      str_detect(raw_item, "reg(ard)?$") ~ "celebrated",
      str_detect(raw_item, "typ$") ~ "typical",
      str_detect(raw_item, "keep$") ~ "job_security",
      str_detect(raw_item, "get$") ~ "job_outcome",
      TRUE ~ NA_character_
    )
  )


# Check to make sure it looks correct and get a count

head(attr_long, 20)
```

```
## # A tibble: 20 × 5
##    responseid        raw_item    choice_raw                    severity attribute
##    <chr>             <chr>       <chr>                         <chr>    <chr>
##  1 R_1kWnphloUO7A7Tj minsmart    neither reported (K) nor re… minor    smarter
##  2 R_1kWnphloUO7A7Tj minbetter   reporting researcher (S)      minor    better_r…
##  3 R_1kWnphloUO7A7Tj mincreative reporting researcher (S)      minor    creative
##  4 R_1kWnphloUO7A7Tj minrather   reporting researcher (S)      minor    rather_be
##  5 R_1kWnphloUO7A7Tj minshould   reporting researcher (S)      minor    should_be
##  6 R_1kWnphloUO7A7Tj minregard   reporting researcher (S)      minor    celebrat…
##  7 R_1kWnphloUO7A7Tj mintyp      neither reported (K) nor re… minor    typical
##  8 R_1kWnphloUO7A7Tj minkeep     reporting researcher (S)      minor    job_secu…
##  9 R_1kWnphloUO7A7Tj minget      reporting researcher (S)      minor    job_outc…
## 10 R_1kWnphloUO7A7Tj modsmart    reporting researcher (F)      moderate smarter
## 11 R_1kWnphloUO7A7Tj modbett     reporting researcher (F)      moderate better_r…
## 12 R_1kWnphloUO7A7Tj modcreat    neither reported (H) nor re… moderate creative
## 13 R_1kWnphloUO7A7Tj modrath     reporting researcher (F)      moderate rather_be
## 14 R_1kWnphloUO7A7Tj modshou     reporting researcher (F)      moderate should_be
## 15 R_1kWnphloUO7A7Tj modreg      reporting researcher (F)      moderate celebrat…
## 16 R_1kWnphloUO7A7Tj modtyp      neither reported (H) nor re… moderate typical
## 17 R_1kWnphloUO7A7Tj modkeep     reporting researcher (F)      moderate job_secu…
## 18 R_1kWnphloUO7A7Tj modget      reporting researcher (F)      moderate job_outc…
## 19 R_1kWnphloUO7A7Tj majsmart    neither reported (X) nor re… major    smarter
## 20 R_1kWnphloUO7A7Tj majbet      reporting researcher (Y)      major    better_r…
```

```
table(attr_long$severity, useNA = "ifany")
```

```
##
##    major    minor moderate
##     4095     4095     4095
```

```
table(attr_long$attribute, useNA = "ifany")
```

```
##
## better_researcher        celebrated          creative       job_outcome
##              1365              1365              1365              1365
##      job_security         rather_be         should_be           smarter
##              1365              1365              1365              1365
##           typical
##              1365
```

```
# Set roles

attr_long2 = attr_long %>%
  tidyr::crossing(
    researcher = c("investigator", "inspector")
  )


# Figure out how many rows for the data

head(attr_long2, 20)
```

```
## # A tibble: 20 × 6
##    responseid        raw_item  choice_raw          severity attribute researcher
##    <chr>             <chr>     <chr>               <chr>    <chr>     <chr>
##  1 R_100RpLnF3upbBKT majbet    reporting research… major    better_r… inspector
##  2 R_100RpLnF3upbBKT majbet    reporting research… major    better_r… investiga…
##  3 R_100RpLnF3upbBKT majcreat  neither reported (… major    creative  inspector
##  4 R_100RpLnF3upbBKT majcreat  neither reported (… major    creative  investiga…
##  5 R_100RpLnF3upbBKT majget    reporting research… major    job_outc… inspector
##  6 R_100RpLnF3upbBKT majget    reporting research… major    job_outc… investiga…
##  7 R_100RpLnF3upbBKT majkeep   reporting research… major    job_secu… inspector
##  8 R_100RpLnF3upbBKT majkeep   reporting research… major    job_secu… investiga…
##  9 R_100RpLnF3upbBKT majrath   reporting research… major    rather_be inspector
## 10 R_100RpLnF3upbBKT majrath   reporting research… major    rather_be investiga…
## 11 R_100RpLnF3upbBKT majreg    reporting research… major    celebrat… inspector
## 12 R_100RpLnF3upbBKT majreg    reporting research… major    celebrat… investiga…
## 13 R_100RpLnF3upbBKT majshould reporting research… major    should_be inspector
## 14 R_100RpLnF3upbBKT majshould reporting research… major    should_be investiga…
## 15 R_100RpLnF3upbBKT majsmart  reporting research… major    smarter   inspector
## 16 R_100RpLnF3upbBKT majsmart  reporting research… major    smarter   investiga…
## 17 R_100RpLnF3upbBKT majtyp    reported researche… major    typical   inspector
## 18 R_100RpLnF3upbBKT majtyp    reported researche… major    typical   investiga…
## 19 R_100RpLnF3upbBKT minbetter neither reported (… minor    better_r… inspector
## 20 R_100RpLnF3upbBKT minbetter neither reported (… minor    better_r… investiga…
```

```
dim(attr_long)
```

```
## [1] 12285     5
```

```
dim(attr_long2)
```

```
## [1] 24570     6
```

```
# Find the correct columns and attribute them to role type

attr_long2 = attr_long2 %>%
  mutate(
    chosen = case_when(
      str_detect(choice_raw, "^reported researcher") &
        researcher == "investigator" ~ 1,
      str_detect(choice_raw, "^reporting researcher") &
        researcher == "inspector" ~ 1,
      str_detect(choice_raw, "^neither") ~ 0,
      TRUE ~ 0
    )
  )


# Check to make sure it looks correct and get a count

head(attr_long2, 20)
```

```
## # A tibble: 20 × 7
##    responseid        raw_item  choice_raw   severity attribute researcher chosen
##    <chr>             <chr>     <chr>        <chr>    <chr>     <chr>       <dbl>
##  1 R_100RpLnF3upbBKT majbet    reporting r… major    better_r… inspector       1
##  2 R_100RpLnF3upbBKT majbet    reporting r… major    better_r… investiga…      0
##  3 R_100RpLnF3upbBKT majcreat  neither rep… major    creative  inspector       0
##  4 R_100RpLnF3upbBKT majcreat  neither rep… major    creative  investiga…      0
##  5 R_100RpLnF3upbBKT majget    reporting r… major    job_outc… inspector       1
##  6 R_100RpLnF3upbBKT majget    reporting r… major    job_outc… investiga…      0
##  7 R_100RpLnF3upbBKT majkeep   reporting r… major    job_secu… inspector       1
##  8 R_100RpLnF3upbBKT majkeep   reporting r… major    job_secu… investiga…      0
##  9 R_100RpLnF3upbBKT majrath   reporting r… major    rather_be inspector       1
## 10 R_100RpLnF3upbBKT majrath   reporting r… major    rather_be investiga…      0
## 11 R_100RpLnF3upbBKT majreg    reporting r… major    celebrat… inspector       1
## 12 R_100RpLnF3upbBKT majreg    reporting r… major    celebrat… investiga…      0
## 13 R_100RpLnF3upbBKT majshould reporting r… major    should_be inspector       1
## 14 R_100RpLnF3upbBKT majshould reporting r… major    should_be investiga…      0
## 15 R_100RpLnF3upbBKT majsmart  reporting r… major    smarter   inspector       1
## 16 R_100RpLnF3upbBKT majsmart  reporting r… major    smarter   investiga…      0
## 17 R_100RpLnF3upbBKT majtyp    reported re… major    typical   inspector       0
## 18 R_100RpLnF3upbBKT majtyp    reported re… major    typical   investiga…      1
## 19 R_100RpLnF3upbBKT minbetter neither rep… minor    better_r… inspector       0
## 20 R_100RpLnF3upbBKT minbetter neither rep… minor    better_r… investiga…      0
```

```
table(attr_long2$chosen, useNA = "ifany")
```

```
##
##     0     1
## 15303  9267
```

```
table(attr_long2$researcher, attr_long2$chosen)
```

```
##
##                    0     1
##   inspector     4731  7554
##   investigator 10572  1713
```

```
# Set up the data for the models

attr_model_dat = attr_long2 %>%
  mutate(
    researcher_ec = if_else(
      researcher == "inspector",
      1,
      -1
    ),
    severity_num = case_when(
      severity == "minor" ~ -1,
      severity == "moderate" ~ 0,
      severity == "major" ~ 1
    ),
    responseid = factor(responseid),
    researcher = factor(
      researcher,
      levels = c("investigator", "inspector")
    ),
    severity = factor(
      severity,
      levels = c("minor", "moderate", "major")
    ),
    attribute = factor(
      attribute,
      levels = c(
        "smarter",
        "better_researcher",
        "creative",
        "rather_be",
        "should_be",
        "celebrated",
        "typical",
        "job_security",
        "job_outcome"
      )
    ),
    vignette_id = factor(severity)
  )


# Create some model tables

str(attr_model_dat)
```

```
## tibble [24,570 × 10] (S3: tbl_df/tbl/data.frame)
##  $ responseid   : Factor w/ 455 levels "R_100RpLnF3upbBKT",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ raw_item     : chr [1:24570] "majbet" "majbet" "majcreat" "majcreat" ...
##  $ choice_raw   : chr [1:24570] "reporting researcher (Y)" "reporting researcher (Y)" "neither reported (X) nor reporting
researcher (Y)" "neither reported (X) nor reporting researcher (Y)" ...
##  $ severity     : Factor w/ 3 levels "minor","moderate",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ attribute    : Factor w/ 9 levels "smarter","better_researcher",..: 2 2 3 3 9 9 8 8 4 4 ...
##  $ researcher   : Factor w/ 2 levels "investigator",..: 2 1 2 1 2 1 2 1 2 1 ...
##  $ chosen       : num [1:24570] 1 0 0 0 1 0 1 0 1 0 ...
##  $ researcher_ec: num [1:24570] 1 -1 1 -1 1 -1 1 -1 1 -1 ...
##  $ severity_num : num [1:24570] 1 1 1 1 1 1 1 1 1 1 ...
##  $ vignette_id  : Factor w/ 3 levels "minor","moderate",..: 3 3 3 3 3 3 3 3 3 3 ...
```

```
table(attr_model_dat$researcher)
```

```
##
## investigator    inspector
##        12285        12285
```

```
table(attr_model_dat$severity)
```

```
##
##    minor moderate    major
##     8190     8190     8190
```

```
table(attr_model_dat$attribute, useNA = "ifany")
```

```
##
##           smarter better_researcher          creative         rather_be
##              2730              2730              2730              2730
##         should_be        celebrated           typical      job_security
##              2730              2730              2730              2730
##       job_outcome
##              2730
```

```
table(attr_model_dat$researcher_ec)
```

```
##
##    -1     1
## 12285 12285
```

```
table(attr_model_dat$severity_num)
```

```
##
##   -1    0    1
## 8190 8190 8190
```

```
# Set attribute comparisons to the grand mean rather than using
# one attribute as the reference category

contrasts(attr_model_dat$attribute) <- contr.sum(
  nlevels(attr_model_dat$attribute)
)

# Check the contrasts

contrasts(attr_model_dat$attribute)
```

```
##                  [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## smarter             1    0    0    0    0    0    0    0
## better_researcher   0    1    0    0    0    0    0    0
## creative            0    0    1    0    0    0    0    0
## rather_be           0    0    0    1    0    0    0    0
## should_be           0    0    0    0    1    0    0    0
## celebrated          0    0    0    0    0    1    0    0
## typical             0    0    0    0    0    0    1    0
## job_security        0    0    0    0    0    0    0    1
## job_outcome        -1   -1   -1   -1   -1   -1   -1   -1
```

```
# Run our first model in the model building for the exploratory analysis

A1 = glmer(
  chosen ~ researcher_ec + attribute +
    (1 | responseid) + (1 | vignette_id),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)


# A2 tests whether choice differs by severity, controlling for attribute

A2 = glmer(
  chosen ~ attribute + severity_num +
    (1 | responseid) + (1 | vignette_id),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 2e5)
  )
)


# A3 tests whether choice differs by BOTH researcher and severity,
# while controlling for attribute

A3 = glmer(
  chosen ~ researcher_ec + attribute + severity_num +
    (1 | responseid) + (1 | vignette_id),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 2e5)
  )
)


# A4 tests whether the effect of researcher depends on severity,
# while controlling for attribute

A4 = glmer(
  chosen ~ researcher_ec * severity_num + attribute +
    (1 | responseid) + (1 | vignette_id),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 2e5)
  )
)


# Summarize models

summary(A1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: chosen ~ researcher_ec + attribute + (1 | responseid) + (1 |
##     vignette_id)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   25172.4   25269.7  -12574.2   25148.4     24558
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0534 -0.5041 -0.3339  0.6316  8.5078
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  responseid  (Intercept) 0.44346  0.66592
##  vignette_id (Intercept) 0.00331  0.05753
## Number of obs: 24570, groups:  responseid, 455; vignette_id, 3
##
## Fixed effects:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.74488    0.04858 -15.335  < 2e-16 ***
## researcher_ec  1.24162    0.01689  73.532  < 2e-16 ***
## attribute1    -0.48165    0.04566 -10.548  < 2e-16 ***
## attribute2     0.09688    0.04424   2.190   0.0285 *
## attribute3    -0.62825    0.04629 -13.573  < 2e-16 ***
## attribute4     0.06369    0.04428   1.438   0.1504
## attribute5     0.17622    0.04415   3.991 6.57e-05 ***
## attribute6     0.22231    0.04411   5.040 4.67e-07 ***
## attribute7    -0.01413    0.04440  -0.318   0.7504
## attribute8     0.30102    0.04406   6.832 8.39e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ attrb1 attrb2 attrb3 attrb4 attrb5 attrb6 attrb7
## researchr_c -0.112
## attribute1   0.014 -0.072
## attribute2  -0.003  0.013 -0.128
## attribute3   0.020 -0.088 -0.126 -0.130
## attribute4  -0.002  0.007 -0.128 -0.123 -0.130
## attribute5  -0.006  0.027 -0.129 -0.122 -0.131 -0.122
## attribute6  -0.007  0.035 -0.129 -0.122 -0.132 -0.122 -0.121
## attribute7   0.000 -0.006 -0.127 -0.123 -0.129 -0.124 -0.123 -0.123
## attribute8  -0.009  0.049 -0.130 -0.121 -0.133 -0.122 -0.120 -0.119 -0.123
```

```
summary(A2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## chosen ~ attribute + severity_num + (1 | responseid) + (1 | vignette_id)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   31858.6   31955.9  -15917.3   31834.6     24558
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.0678 -0.8111 -0.6071  1.1062  2.9325
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  responseid  (Intercept) 0.2209   0.47
##  vignette_id (Intercept) 0.0000   0.00
## Number of obs: 24570, groups:  responseid, 455; vignette_id, 3
##
## Fixed effects:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.540510   0.025877 -20.888  < 2e-16 ***
## attribute1   -0.356136   0.039691  -8.973  < 2e-16 ***
## attribute2    0.073445   0.037569   1.955 0.050589 .
## attribute3   -0.468925   0.040504 -11.577  < 2e-16 ***
## attribute4    0.049342   0.037649   1.311 0.189999
## attribute5    0.130858   0.037397   3.499 0.000467 ***
## attribute6    0.164089   0.037309   4.398 1.09e-05 ***
## attribute7   -0.007371   0.037855  -0.195 0.845605
## attribute8    0.220684   0.037179   5.936 2.93e-09 ***
## severity_num  0.048734   0.016446   2.963 0.003043 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) attrb1 attrb2 attrb3 attrb4 attrb5 attrb6 attrb7 attrb8
## attribute1   0.018
## attribute2  -0.006 -0.130
## attribute3   0.027 -0.141 -0.134
## attribute4  -0.005 -0.130 -0.121 -0.134
## attribute5  -0.008 -0.129 -0.120 -0.133 -0.120
## attribute6  -0.009 -0.129 -0.120 -0.133 -0.120 -0.119
## attribute7  -0.002 -0.131 -0.122 -0.134 -0.122 -0.121 -0.121
## attribute8  -0.010 -0.129 -0.119 -0.132 -0.119 -0.118 -0.118 -0.120
## severity_nm -0.005 -0.002  0.000 -0.003  0.000  0.001  0.001  0.000  0.001
```

```
summary(A3)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## chosen ~ researcher_ec + attribute + severity_num + (1 | responseid) +
##     (1 | vignette_id)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   25169.9   25275.4  -12572.0   25143.9     24557
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0794 -0.5026 -0.3330  0.6344  8.5045
##
## Random effects:
##  Groups      Name        Variance  Std.Dev.
##  responseid  (Intercept) 4.427e-01 6.654e-01
##  vignette_id (Intercept) 1.917e-14 1.385e-07
## Number of obs: 24570, groups:  responseid, 455; vignette_id, 3
##
## Fixed effects:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.74486    0.03542 -21.028  < 2e-16 ***
## researcher_ec  1.24162    0.01689  73.519  < 2e-16 ***
## attribute1    -0.48165    0.04567 -10.547  < 2e-16 ***
## attribute2     0.09688    0.04424   2.190 0.028531 *
## attribute3    -0.62824    0.04629 -13.572  < 2e-16 ***
## attribute4     0.06369    0.04428   1.438 0.150390
## attribute5     0.17622    0.04415   3.991 6.58e-05 ***
## attribute6     0.22231    0.04411   5.039 4.67e-07 ***
## attribute7    -0.01413    0.04440  -0.318 0.750381
## attribute8     0.30102    0.04406   6.831 8.40e-12 ***
## severity_num   0.06656    0.01928   3.452 0.000557 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ attrb1 attrb2 attrb3 attrb4 attrb5 attrb6 attrb7
## researchr_c -0.153
## attribute1   0.020 -0.072
## attribute2  -0.005  0.013 -0.128
## attribute3   0.027 -0.088 -0.126 -0.130
## attribute4  -0.003  0.007 -0.128 -0.123 -0.130
## attribute5  -0.008  0.027 -0.129 -0.122 -0.131 -0.122
## attribute6  -0.009  0.035 -0.129 -0.122 -0.132 -0.122 -0.121
## attribute7   0.000 -0.006 -0.127 -0.123 -0.129 -0.124 -0.123 -0.123
## attribute8  -0.012  0.049 -0.130 -0.121 -0.133 -0.122 -0.120 -0.119 -0.123
## severity_nm -0.006  0.025 -0.003  0.001 -0.004  0.001  0.001  0.002  0.000
##             attrb8
## researchr_c
## attribute1
## attribute2
## attribute3
## attribute4
## attribute5
## attribute6
## attribute7
## attribute8
## severity_nm  0.002
```

```
summary(A4)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula:
## chosen ~ researcher_ec * severity_num + attribute + (1 | responseid) +
##     (1 | vignette_id)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   24713.6   24827.2  -12342.8   24685.6     24556
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4484 -0.5169 -0.2974  0.6220  8.6407
##
## Random effects:
##  Groups      Name        Variance  Std.Dev.
##  responseid  (Intercept) 0.4652766 0.68211
##  vignette_id (Intercept) 0.0006473 0.02544
## Number of obs: 24570, groups:  responseid, 455; vignette_id, 3
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.77126    0.03927 -19.642  < 2e-16 ***
## researcher_ec               1.27804    0.01753  72.896  < 2e-16 ***
## severity_num               -0.06793    0.02772  -2.450   0.0143 *
## attribute1                 -0.49156    0.04613 -10.656  < 2e-16 ***
## attribute2                  0.09881    0.04469   2.211   0.0270 *
## attribute3                 -0.64110    0.04677 -13.709  < 2e-16 ***
## attribute4                  0.06491    0.04473   1.451   0.1467
## attribute5                  0.17983    0.04461   4.031 5.55e-05 ***
## attribute6                  0.22691    0.04457   5.091 3.57e-07 ***
## attribute7                 -0.01452    0.04485  -0.324   0.7462
## attribute8                  0.30733    0.04453   6.902 5.14e-12 ***
## researcher_ec:severity_num  0.44211    0.02107  20.981  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ svrty_ attrb1 attrb2 attrb3 attrb4 attrb5 attrb6
## researchr_c -0.158
## severity_nm  0.068 -0.119
## attribute1   0.018 -0.072 -0.004
## attribute2  -0.004  0.013  0.002 -0.128
## attribute3   0.025 -0.088 -0.006 -0.126 -0.130
## attribute4  -0.003  0.007  0.002 -0.128 -0.123 -0.130
## attribute5  -0.007  0.027  0.002 -0.129 -0.122 -0.131 -0.122
## attribute6  -0.009  0.035  0.002 -0.129 -0.122 -0.132 -0.122 -0.121
## attribute7   0.000 -0.006  0.002 -0.127 -0.123 -0.129 -0.124 -0.123 -0.123
## attribute8  -0.012  0.049  0.001 -0.130 -0.121 -0.133 -0.122 -0.120 -0.119
## rsrchr_c:s_ -0.075  0.215 -0.257 -0.017  0.003 -0.022  0.002  0.006  0.008
##             attrb7 attrb8
## researchr_c
## severity_nm
## attribute1
## attribute2
## attribute3
## attribute4
## attribute5
## attribute6
## attribute7
## attribute8  -0.123
## rsrchr_c:s_ -0.001  0.012
```

```
# Get the number and percentage of participants who chose
# each researcher for each attribute and vignette

attribute_descriptives = attr_model_dat %>%
  group_by(vignette_id, severity_num, attribute, researcher_ec) %>%
  summarise(
    N = sum(chosen == 1, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(vignette_id, severity_num, attribute) %>%
  mutate(
    Total_N = sum(N),
    Percent = round(N / Total_N * 100, 2)
  ) %>%
  tidyr::pivot_wider(
    names_from = researcher_ec,
    values_from = c(N, Percent),
    values_fill = 0,
    names_glue = "{.value}_{researcher_ec}"
  ) %>%
  rename(
    Investigator_N = `N_-1`,
    Inspector_N = `N_1`,
    Investigator_Percent = `Percent_-1`,
    Inspector_Percent = `Percent_1`
  ) %>%
  ungroup()

print(attribute_descriptives, n = 27)
```

```
## # A tibble: 27 × 8
##    vignette_id severity_num attribute         Total_N Investigator_N Inspector_N
##    <fct>              <dbl> <fct>               <int>          <int>       <int>
##  1 minor                 -1 smarter               248             59         189
##  2 minor                 -1 better_researcher     337             48         289
##  3 minor                 -1 creative              255            144         111
##  4 minor                 -1 rather_be             340             70         270
##  5 minor                 -1 should_be             346             42         304
##  6 minor                 -1 celebrated            362             82         280
##  7 minor                 -1 typical               358            213         145
##  8 minor                 -1 job_security          365             69         296
##  9 minor                 -1 job_outcome           367            107         260
## 10 moderate               0 smarter               269             31         238
## 11 moderate               0 better_researcher     361             25         336
## 12 moderate               0 creative              253            146         107
## 13 moderate               0 rather_be             357             20         337
## 14 moderate               0 should_be             380             11         369
## 15 moderate               0 celebrated            387             43         344
## 16 moderate               0 typical               340            146         194
## 17 moderate               0 job_security          400             31         369
## 18 moderate               0 job_outcome           386             55         331
## 19 major                  1 smarter               300             16         284
## 20 major                  1 better_researcher     373             13         360
## 21 major                  1 creative              248            124         124
## 22 major                  1 rather_be             359             15         344
## 23 major                  1 should_be             381             11         370
## 24 major                  1 celebrated            379             27         352
## 25 major                  1 typical               323            112         211
## 26 major                  1 job_security          399             19         380
## 27 major                  1 job_outcome           394             34         360
## # ℹ 2 more variables: Investigator_Percent <dbl>, Inspector_Percent <dbl>
```

```
### 5. Overall attribute test + pairwise comparisons

# Test the overall effect of attribute

anova(A4, test = "Chisq")
```

```
## Warning in anova.merMod(A4, test = "Chisq"): additional arguments ignored:
## 'test'
```

```
## Analysis of Variance Table
##                            npar Sum Sq Mean Sq   F value
## researcher_ec                 1 4701.0  4701.0 4700.9783
## severity_num                  1    7.5     7.5    7.4613
## attribute                     8  354.3    44.3   44.2830
## researcher_ec:severity_num    1  440.2   440.2  440.1845
```

```
# Get estimated marginal means for each attribute

library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
attribute_emmeans = emmeans(
  A4,
  ~ attribute,
  type = "response"
)

print(attribute_emmeans)
```

```
##  attribute          prob      SE  df asymp.LCL asymp.UCL
##  smarter           0.220 0.01050 Inf     0.201     0.242
##  better_researcher 0.338 0.01330 Inf     0.312     0.364
##  creative          0.196 0.00973 Inf     0.177     0.216
##  rather_be         0.330 0.01310 Inf     0.305     0.357
##  should_be         0.356 0.01360 Inf     0.330     0.383
##  celebrated        0.367 0.01370 Inf     0.341     0.394
##  typical           0.313 0.01280 Inf     0.289     0.339
##  job_security      0.386 0.01400 Inf     0.359     0.414
##  job_outcome       0.377 0.01390 Inf     0.350     0.405
##
## Results are averaged over the levels of: researcher_ec
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```
# Compare attributes to each other

pairs(
  attribute_emmeans,
  adjust = "bonferroni"
)
```

```
##  contrast                         odds.ratio     SE  df null z.ratio p.value
##  smarter / better_researcher           0.554 0.0378 Inf    1  -8.654 <0.0001
##  smarter / creative                    1.161 0.0809 Inf    1   2.146  1.0000
##  smarter / rather_be                   0.573 0.0391 Inf    1  -8.154 <0.0001
##  smarter / should_be                   0.511 0.0348 Inf    1  -9.847 <0.0001
##  smarter / celebrated                  0.487 0.0332 Inf    1 -10.540 <0.0001
##  smarter / typical                     0.621 0.0424 Inf    1  -6.984 <0.0001
##  smarter / job_security                0.450 0.0307 Inf    1 -11.721 <0.0001
##  smarter / job_outcome                 0.467 0.0318 Inf    1 -11.164 <0.0001
##  better_researcher / creative          2.096 0.1440 Inf    1  10.760 <0.0001
##  better_researcher / rather_be         1.034 0.0693 Inf    1   0.506  1.0000
##  better_researcher / should_be         0.922 0.0617 Inf    1  -1.211  1.0000
##  better_researcher / celebrated        0.880 0.0588 Inf    1  -1.916  1.0000
##  better_researcher / typical           1.120 0.0752 Inf    1   1.689  1.0000
##  better_researcher / job_security      0.812 0.0542 Inf    1  -3.122  0.0648
##  better_researcher / job_outcome       0.843 0.0563 Inf    1  -2.553  0.3846
##  creative / rather_be                  0.494 0.0340 Inf    1 -10.264 <0.0001
##  creative / should_be                  0.440 0.0302 Inf    1 -11.943 <0.0001
##  creative / celebrated                 0.420 0.0289 Inf    1 -12.630 <0.0001
##  creative / typical                    0.534 0.0368 Inf    1  -9.101 <0.0001
##  creative / job_security               0.387 0.0266 Inf    1 -13.800 <0.0001
##  creative / job_outcome                0.402 0.0277 Inf    1 -13.248 <0.0001
##  rather_be / should_be                 0.891 0.0597 Inf    1  -1.717  1.0000
##  rather_be / celebrated                0.850 0.0569 Inf    1  -2.422  0.5561
##  rather_be / typical                   1.083 0.0727 Inf    1   1.183  1.0000
##  rather_be / job_security              0.785 0.0525 Inf    1  -3.627  0.0103
##  rather_be / job_outcome               0.815 0.0545 Inf    1  -3.058  0.0802
##  should_be / celebrated                0.954 0.0637 Inf    1  -0.705  1.0000
##  should_be / typical                   1.215 0.0814 Inf    1   2.899  0.1347
##  should_be / job_security              0.880 0.0587 Inf    1  -1.911  1.0000
##  should_be / job_outcome               0.914 0.0610 Inf    1  -1.342  1.0000
##  celebrated / typical                  1.273 0.0853 Inf    1   3.603  0.0113
##  celebrated / job_security             0.923 0.0615 Inf    1  -1.207  1.0000
##  celebrated / job_outcome              0.958 0.0639 Inf    1  -0.637  1.0000
##  typical / job_security                0.725 0.0485 Inf    1  -4.806 <0.0001
##  typical / job_outcome                 0.753 0.0504 Inf    1  -4.239  0.0008
##  job_security / job_outcome            1.039 0.0692 Inf    1   0.569  1.0000
##
## Results are averaged over the levels of: researcher_ec
## P value adjustment: bonferroni method for 36 tests
## Tests are performed on the log odds ratio scale
```

Integrity & LIKEABILITY ANALYSIS (RQ1, RQ2) + TRUENESS (RQ3)

Research Question 1: How is integrity influenced by role and QRP severity? • Hypothesis 1a: Inspectors will be rated higher in integrity than investigators. • Hypothesis 1b: The relative integrity of inspectors versus investigators will increase as QRP severity increases.

Research Question 2: How is likeability influenced by role and QRP severity? • Hypothesis 2a: Investigators will be rated more likable than inspectors. • Hypothesis 2b: At low QRP severity, investigators will be rated as more likeable than inspectors, but at high QRP severity, inspectors will be rated as more likeable than investigators.

Research Question 3: Is the perceived trueness of the finding influenced by QRP severity? • Hypothesis 3: Perceived trueness will decrease as QRP severity increases.

Research Questions 1–2: Integrity and Likeability Step-up MLMs will test Role, Severity, and Role × Severity interaction:

Model 1: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + v0k + μ0jk + εijk Model 2: Integrity/Likeabilityijk = γ000 + γ010 (Severityk) + v0k + μ0k + εijk Model 3: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + γ010 (Severityk) + v0k + μ0jk + εijk Model 4: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + γ010 (Severityk) + γ110(Researcherj × Severityk) + v0k + μ0jk + εijk

Research Question 3: Trueness of Findings Model FT1: Truenessijk= γ000 + γ010 (Severityk) + v0k + μ0j + εijk

Build dataset

```
# Vignette metadata
# Each vignette pair: investigator (inv) and inspector (insp)
# Severity:minor = vignette K/S, moderate = H/F, major = X/Y
# Integrity items: hon, sin, eth  (honest, sincere, ethical)
# Likeability: like, app (likeable, approachable)
# Trueness (conf): investigator only, 1 item per vignette

vignette_map = list(
  list(vig_letter = "k", vig_id = "minor",    role = "investigator",
       hon = "khon", sin = "ksin",  eth = "keth",
       like = "klike",  app = "kapp",  conf = "kcon"),
  list(vig_letter = "s", vig_id = "minor",    role = "inspector",
       hon = "shon",    sin = "ssin",  eth = "seth",
       like = "slike",  app = "sapp",  conf = NA),
  list(vig_letter = "h", vig_id = "moderate", role = "investigator",
       hon = "hhon",    sin = "hsin",  eth = "heth",
       like = "hlike",  app = "happ",  conf = "hcon"),
  list(vig_letter = "f", vig_id = "moderate", role = "inspector",
       hon = "fhon",    sin = "fsin",  eth = "feth",
       like = "flike",  app = "fapp",  conf = NA),
  list(vig_letter = "x", vig_id = "major",    role = "investigator",
       hon = "xhon",   sin = "xsin",  eth = "xeth",
       like = "xlike",  app = "xapp",  conf = "xcon"),
  list(vig_letter = "y", vig_id = "major",    role = "inspector",
       hon = "yhon",    sin = "ysin",  eth = "yeth",
       like = "ylike",  app = "yapp",  conf = NA)
)
# vig_letter is the TRUE vignette identifier (6 distinct narrative vignettes:
# k/s/h/f/x/y). vig_id (severity) is NOT a valid vignette-level random effect
# on its own — it only has 3 levels and is perfectly confounded with
# severity_num, the key fixed effect. vig_letter fixes this: it captures
# idiosyncratic vignette-writing variance independent of role and severity.

#Helper: extract numeric value from labelled string
parse_num = function(x) as.numeric(str_extract(as.character(x), "^\\d"))

#Build one row per participant × vignette × role
tl_rows = bind_rows(lapply(vignette_map, function(v) {
  filtered_by_age %>%                          # use cleaned dataset
    transmute(
      responseid,
      vig_letter = v$vig_letter,
      vig_id   = v$vig_id,
      role     = v$role,
      hon_raw  = parse_num(.data[[v$hon]]),
      sin_raw  = parse_num(.data[[v$sin]]),
      eth_raw  = parse_num(.data[[v$eth]]),
      like_raw = parse_num(.data[[v$like]]),
      app_raw  = parse_num(.data[[v$app]]),
      conf_raw = if (!is.na(v$conf)) parse_num(.data[[v$conf]]) else NA_real_
    )
}))

#Composites & factors
tl_dat = tl_rows %>%
  mutate(
    Integrity      = rowMeans(cbind(hon_raw, sin_raw, eth_raw), na.rm = TRUE),
    likeability = rowMeans(cbind(like_raw, app_raw), na.rm = TRUE),
    severity   = factor(vig_id,   levels = c("minor", "moderate", "major")),
    role       = factor(role,     levels = c("investigator", "inspector")),
    responseid = factor(responseid),
    vignette_id = factor(vig_letter, levels = c("k","s","h","f","x","y")),
    # effect-coded predictors
    role_ec    = if_else(role == "inspector", 1, -1),
    severity_num = case_when(
      severity == "minor"    ~ -1,
      severity == "moderate" ~  0,
      severity == "major"    ~  1
```

```
    )
  )

glimpse(tl_dat)
```

```
## Rows: 2,730
## Columns: 16
## $ responseid   <fct> R_1kWnphloUO7A7Tj, R_5lX85aTYV0koeHL, R_1fs9EL5QZj1yBtB, …
## $ vig_letter   <chr> "k", "k", "k", "k", "k", "k", "k", "k", "k", "k", "k", "k…
## $ vig_id       <chr> "minor", "minor", "minor", "minor", "minor", "minor", "mi…
## $ role         <fct> investigator, investigator, investigator, investigator, i…
## $ hon_raw      <dbl> 3, 5, 2, 4, 3, 2, 3, 2, 5, 3, 2, 2, 2, 2, 4, 2, 4, 2, 4, …
## $ sin_raw      <dbl> 2, 5, 2, 4, 2, 4, 4, 2, 5, 4, 2, 4, 3, 2, 4, 3, 4, 3, 2, …
## $ eth_raw      <dbl> 1, 5, 3, 2, 4, 4, 2, 2, 5, 4, 1, 2, 2, 1, 4, 3, 3, 2, 2, …
## $ like_raw     <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 5, 2, 2, 4, 4, 3, 3, 3, 3, 3, 3, …
## $ app_raw      <dbl> 2, 4, 3, 5, 3, 4, 4, 4, 5, 4, 2, 4, 4, 4, 2, 4, 3, 3, 2, …
## $ conf_raw     <dbl> 3, 3, 3, 4, 4, 3, 3, 3, 4, 3, 2, 4, 3, 2, 3, 4, 3, 4, 2, …
## $ Integrity    <dbl> 2.000000, 5.000000, 2.333333, 3.333333, 3.000000, 3.33333…
## $ likeability  <dbl> 2.5, 3.5, 3.0, 4.0, 3.0, 3.5, 3.5, 3.5, 5.0, 3.0, 2.0, 4.…
## $ severity     <fct> minor, minor, minor, minor, minor, minor, minor, minor, m…
## $ vignette_id  <fct> k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, …
## $ role_ec      <dbl> -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -…
## $ severity_num <dbl> -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -…
```

Check structure and counts

```
table(tl_dat$role, tl_dat$severity)
```

```
##
##                minor moderate major
##   investigator   455      455   455
##   inspector      455      455   455
```

```
# Quick descriptives
tl_dat %>%
  group_by(role, severity) %>%
  summarise(
    n          = sum(!is.na(Integrity)),
    Integrity_M    = round(mean(Integrity, na.rm = TRUE), 2),
    Integrity_SD   = round(sd(Integrity,   na.rm = TRUE), 2),
    like_M     = round(mean(likeability, na.rm = TRUE), 2),
    like_SD    = round(sd(likeability,   na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  kable(caption = "Descriptives: Integrity & Likeability by Role × Severity") %>%
  style_table()
```

Descriptives: Integrity & Likeability by Role × Severity

role	severity	n	Integrity_M	Integrity_SD	like_M	like_SD
investigator	minor	446	2.79	0.86	3.03	0.77
investigator	moderate	447	1.69	0.73	2.36	0.81
investigator	major	447	1.39	0.70	2.02	0.84
inspector	minor	445	4.17	0.83	3.59	0.94
inspector	moderate	447	4.35	0.73	3.87	0.86
inspector	major	446	4.40	0.73	3.94	0.86

##Reliability & validation checks

```
library(psych)

#Integrity alpha per role × severity (using the 3 items)
Integrity_items_inv = filtered_by_age %>%
  transmute(
    min_hon = parse_num(khon), min_sin = parse_num(ksin),   min_eth = parse_num(keth),
    mod_hon = parse_num(hhon),    mod_sin = parse_num(hsin),   mod_eth = parse_num(heth),
    maj_hon = parse_num(xhon),   maj_sin = parse_num(xsin),   maj_eth = parse_num(xeth)
  )

Integrity_items_insp = filtered_by_age %>%
  transmute(
    min_hon = parse_num(shon),  min_sin = parse_num(ssin),  min_eth = parse_num(seth),
    mod_hon = parse_num(fhon),  mod_sin = parse_num(fsin),  mod_eth = parse_num(feth),
    maj_hon = parse_num(yhon),  maj_sin = parse_num(ysin),  maj_eth = parse_num(yeth)
  )

psych::alpha(Integrity_items_inv,  check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.7740442 0.7812991
```

```
psych::alpha(Integrity_items_insp, check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.8659431 0.8674587
```

```
#Likeability alpha
like_items_inv = filtered_by_age %>%
  transmute(
    min_like = parse_num(klike), min_app = parse_num(kapp),
    mod_like = parse_num(hlike), mod_app = parse_num(happ),
    maj_like = parse_num(xlike), maj_app = parse_num(xapp)
  )

like_items_insp = filtered_by_age %>%
  transmute(
    min_like = parse_num(slike), min_app = parse_num(sapp),
    mod_like = parse_num(flike), mod_app = parse_num(fapp),
    maj_like = parse_num(ylike), maj_app = parse_num(yapp)
  )

psych::alpha(like_items_inv,  check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.7502138 0.7481344
```

```
psych::alpha(like_items_insp, check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.8304724  0.830882
```

```
#Inter-composite correlation
cor.test(tl_dat$Integrity, tl_dat$likeability, use = "complete.obs") %>% tidy() %>% print()
```

```
## # A tibble: 1 × 8
##   estimate statistic p.value parameter conf.low conf.high method     alternative
##      <dbl>     <dbl>   <dbl>     <int>    <dbl>     <dbl> <chr>      <chr>
## 1    0.803      69.8       0      2676    0.789     0.816 Pearson's… two.sided
```

##Model-building strategy

The analyses below use the preregistered step-up structure for each outcome. Models are fit using maximum likelihood (REML = FALSE) so that fixed-effect models can be compared with likelihood-ratio tests. Random slopes are evaluated first for the fully specified interaction model and are stepped down only when the more complex structure is singular. The resulting random-effects structure is then held constant across Models 0–4 so that the likelihood-ratio comparisons test changes in the fixed-effects structure rather than changes in the random-effects structure.

Model sequence for Integrity and Likeability

- Model 0: Null model
- Model 1: Role
- Model 2: QRP Severity
- Model 3: Role + QRP Severity
- Model 4: Role × QRP Severity

Model 1 tests the main effect of researcher role, corresponding to H1a for Integrity and H2a for Likeability. Model 4 tests the Role × QRP Severity interaction, corresponding to H1b and H2b.

```r
library(lme4)
library(lmerTest)
library(emmeans)
library(knitr)
library(kableExtra)

# Extract fixed effects, including Satterthwaite df, t, and p.
get_fixed_effects = function(model) {
  cf = summary(model)$coefficients

  data.frame(
    Parameter = rownames(cf),
    b = cf[, "Estimate"],
    SE = cf[, "Std. Error"],
    df = cf[, "df"],
    t = cf[, "t value"],
    p = cf[, "Pr(>|t|)"],
    row.names = NULL,
    check.names = FALSE
  )
}

# APA-friendly p-value formatting.
fmt_p = function(p) {
  if (is.na(p)) return("")
  if (p < .001) return("< .001")
  formatC(p, digits = 3, format = "f")
}

# APA-friendly number formatting.
fmt_num = function(x, digits = 2) {
  if (length(x) == 0 || is.na(x)) return("")
  formatC(x, digits = digits, format = "f")
}

# Rename model terms for tables.
relabel_terms = function(x) {
  dplyr::recode(
    x,
    "(Intercept)" = "Intercept",
    "role_ec" = "Role",
    "severity_num" = "QRP Severity",
    "role_ec:severity_num" = "Role × QRP Severity",
    .default = x
  )
}

# Likelihood-ratio comparison of two nested ML models.
compare_models = function(previous, current) {
  lr = anova(previous, current)
  data.frame(
    Chisq = lr$Chisq[2],
    df = lr$Df[2],
    p = lr$`Pr(>Chisq)`[2]
  )
}

# Build a model-fit table. Each chi-square is the LR test against
# the immediately preceding model.
model_fit_table = function(models, names) {
  out = lapply(seq_along(models), function(i) {
    m = models[[i]]

    if (i == 1) {
      chisq = NA_real_
      df = NA_real_
      p = NA_real_
    } else {
      lr = compare_models(models[[i - 1]], m)
      chisq = lr$Chisq
```

```
      df = lr$df
      p = lr$p
    }

    data.frame(
      Model = names[i],
      AIC = AIC(m),
      BIC = BIC(m),
      logLik = as.numeric(logLik(m)),
      Chisq = chisq,
      df = df,
      p = p,
      check.names = FALSE
    )
  })

  dplyr::bind_rows(out) %>%
    mutate(
      AIC = round(AIC, 1),
      BIC = round(BIC, 1),
      logLik = round(logLik, 1),
      Chisq = ifelse(is.na(Chisq), "—", sprintf("%.2f", Chisq)),
      df = ifelse(is.na(df), "—", as.character(df)),
      p = ifelse(is.na(p), "—", vapply(p, fmt_p, character(1)))
    )
}

# Extract variance components.
get_variance_components = function(model) {
  vc = as.data.frame(VarCorr(model))
  vc %>%
    select(grp, vcov, sdcor) %>%
    mutate(
      Variance = round(vcov, 3),
      SD = round(sdcor, 3)
    ) %>%
    select(grp, Variance, SD)
}

# ICC for random-intercept components. Random slopes/covariances are not
# interpreted as simple ICC components here.
get_random_intercept_icc = function(model) {
  vc = as.data.frame(VarCorr(model))
  intercepts = vc %>%
    filter(is.na(var1) | var1 == "(Intercept)")

  residual = vc$vcov[vc$grp == "Residual"]
  total = sum(intercepts$vcov) + residual

  intercepts %>%
    mutate(ICC = round(vcov / total, 3)) %>%
    select(grp, ICC)
}
```

##Integrity

Research Question 1: How is integrity influenced by role and QRP severity?

Hypothesis 1a: Inspectors will be rated higher in integrity than investigators.

Hypothesis 1b: The relative integrity of inspectors versus investigators will increase as QRP severity increases.

###Random-effects structure for Integrity

```
T4_max1 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
T4_max1_singular = isSingular(T4_max1, tol = 1e-4)
print(T4_max1_singular)
```

```
## [1] TRUE
```

```
# Step down: remove random interaction slope

T4_max2 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec + severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
T4_max2_singular = isSingular(T4_max2, tol = 1e-4)
print(T4_max2_singular)
```

```
## [1] TRUE
```

```
# Step down: retain random role slope only

T4_max3 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

T4_max3_singular = isSingular(T4_max3, tol = 1e-4)
print(T4_max3_singular)
```

```
## [1] FALSE
```

```
# Test preregistered vignette random intercept with the winning

T4_prereg = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec | responseid) +
    (1 | vignette_id),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

T4_prereg_singular = isSingular(T4_prereg, tol = 1e-4)
print(T4_prereg_singular)
```

```
## [1] FALSE
```

```
# Decision rule used throughout the analysis:
# retain the vignette intercept if the preregistered structure is
# non-singular; otherwise use the participant-only structure.
Integrity_uses_vignette_term = !T4_prereg_singular

cat(
  "Integrity final random-effects structure includes vignette intercept:",
  Integrity_uses_vignette_term,
  "\n"
)
```

```
## Integrity final random-effects structure includes vignette intercept: TRUE
```

###step-up models

```
if (Integrity_uses_vignette_term) {

  random_part_Integrity = "vignette"

  M0_Integrity = lmer(
    Integrity ~ 1 +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T1 = lmer(
    Integrity ~ role_ec +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T2 = lmer(
    Integrity ~ severity_num +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T3 = lmer(
    Integrity ~ role_ec + severity_num +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T4 = T4_prereg

} else {

  random_part_Integrity = "participant_only"

  M0_Integrity = lmer(
    Integrity ~ 1 +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T1 = lmer(
    Integrity ~ role_ec +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T2 = lmer(
    Integrity ~ severity_num +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
```

```
  T3 = lmer(
    Integrity ~ role_ec + severity_num +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T4 = T4_max3
}


# Singularity checks for final models.
cat("Integrity final model singularity:\n")
```

```
## Integrity final model singularity:
```

```
print(isSingular(M0_Integrity))
```

```
## [1] FALSE
```

```
print(isSingular(T1))
```

```
## [1] FALSE
```

```
print(isSingular(T2))
```

```
## [1] FALSE
```

```
print(isSingular(T3))
```

```
## [1] FALSE
```

```
print(isSingular(T4))
```

```
## [1] FALSE
```

```
# Complete step-up comparison, including Model 0.
integrity_models = list(
  M0_Integrity,
  T1,
  T2,
  T3,
  T4
)

integrity_model_names = c(
  "Model 0: Null",
  "Model 1: Role",
  "Model 2: QRP Severity",
  "Model 3: Role + QRP Severity",
  "Model 4: Role × QRP Severity"
)

integrity_fit = model_fit_table(
  integrity_models,
  integrity_model_names
)

print(integrity_fit)
```

```
##                          Model    AIC    BIC  logLik Chisq df      p
## 1                Model 0: Null 5826.3 5861.6 -2907.1     –  –      –
## 2                Model 1: Role 5815.5 5856.7 -2900.7 12.80  1 < .001
## 3        Model 2: QRP Severity 5828.1 5869.3 -2907.0  0.00  0      –
## 4 Model 3: Role + QRP Severity 5815.3 5862.5 -2899.7 14.72  1 < .001
## 5 Model 4: Role × QRP Severity 5805.8 5858.9 -2893.9 11.52  1 < .001
```

```
# Full model summaries.
summary(T1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec + (1 + role_ec | responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5815.5    5856.7   -2900.7    5801.5      2671
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9305 -0.5336  0.0050  0.4975  5.4349
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 0.1850   0.4301
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##             Estimate Std. Error     df t value Pr(>|t|)
## (Intercept)   3.1306     0.1762 6.0354  17.764 1.93e-06 ***
## role_ec       1.1764     0.1770 6.1462   6.645 0.000508 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##         (Intr)
## role_ec 0.002
```

```
summary(T2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ severity_num + (1 + role_ec | responseid) + (1 |
##     vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5828.1    5869.3   -2907.0    5814.1      2671
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9298 -0.5329  0.0068  0.4982  5.4377
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 1.5090   1.2284
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1303     0.5017  5.9881   6.239 0.000791 ***
## severity_num  -0.2891     0.6144  5.9837  -0.471 0.654622
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
summary(T3)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec + severity_num + (1 + role_ec | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5815.3    5862.5   -2899.7    5799.3      2670
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9304 -0.5339  0.0084  0.5009  5.4362
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 0.1292   0.3595
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1306     0.1476  6.0505  21.217 6.54e-07 ***
## role_ec        1.1764     0.1485  6.2091   7.921 0.000181 ***
## severity_num  -0.2891     0.1803  5.9997  -1.603 0.160023
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c
## role_ec     0.003
## severity_nm 0.000  0.000
```

```
summary(T4)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec * severity_num + (1 + role_ec | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5805.8    5858.9   -2893.9    5787.8      2669
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9277 -0.5336  0.0115  0.4970  5.4488
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.04098  0.2024
##              role_ec     0.16783  0.4097   0.32
##  vignette_id (Intercept) 0.01819  0.1349
##  Residual                0.37551  0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           3.13057    0.05713  6.33858  54.794 1.00e-09 ***
## role_ec               1.17638    0.05956  7.48480  19.750 9.94e-08 ***
## severity_num         -0.28909    0.06899  5.98853  -4.191  0.00577 **
## role_ec:severity_num  0.40805    0.06899  5.98864   5.915  0.00105 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c svrty_
## role_ec     0.017
## severity_nm 0.000  0.000
## rl_c:svrty_ 0.000  0.000  0.000
```

###fixed effects and planned role contrasts

```
# Fixed effects from every model, including Satterthwaite df, t, and p.
integrity_fixed_effects = lapply(
  integrity_models,
  get_fixed_effects
)

names(integrity_fixed_effects) = paste0("Model_", 0:4)

print(integrity_fixed_effects)
```

```
## $Model_0
##     Parameter        b        SE       df        t            p
## 1 (Intercept) 3.130303 0.5109191 5.989118 6.126807 0.0008699531
##
## $Model_1
##     Parameter        b        SE       df         t            p
## 1 (Intercept) 3.130567 0.1762325 6.035446 17.763839 1.933673e-06
## 2     role_ec 1.176382 0.1770355 6.146161  6.644893 5.076029e-04
##
## $Model_2
##      Parameter          b        SE       df          t            p
## 1  (Intercept)  3.1302929 0.5017281 5.988079  6.2390223 0.0007910276
## 2 severity_num -0.2890974 0.6143771 5.983693 -0.4705537 0.6546220311
##
## $Model_3
##      Parameter          b        SE       df         t            p
## 1  (Intercept)  3.1305677 0.1475504 6.050460 21.216941 6.544313e-07
## 2      role_ec  1.1763829 0.1485084 6.209082  7.921321 1.809007e-04
## 3 severity_num -0.2890992 0.1803309 5.999654 -1.603159 1.600230e-01
##
## $Model_4
##              Parameter          b         SE       df         t            p
## 1          (Intercept)  3.1305668 0.05713328 6.338584 54.794106 1.004539e-09
## 2              role_ec  1.1763834 0.05956286 7.484802 19.750285 9.935063e-08
## 3         severity_num -0.2890927 0.06898510 5.988534 -4.190654 5.769390e-03
## 4 role_ec:severity_num  0.4080529 0.06898539 5.988637  5.915062 1.046426e-03
```

```
# Planned comparison: inspectors versus investigators at each severity.
# Bonferroni correction is applied across the three severity-specific
# role comparisons.
emm_Integrity = emmeans(
  T4,
  ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1))
)

emm_Integrity_role = emmeans(
  T4,
  pairwise ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_Integrity_role$contrasts)
```

```
## severity_num = -1:
##  contrast              estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -1.54 0.308 16.7  -4.997  0.0001
##
## severity_num =  0:
##  contrast              estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -2.35 0.197 16.1 -11.956 <0.0001
##
## severity_num =  1:
##  contrast              estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -3.17 0.307 16.7 -10.305 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

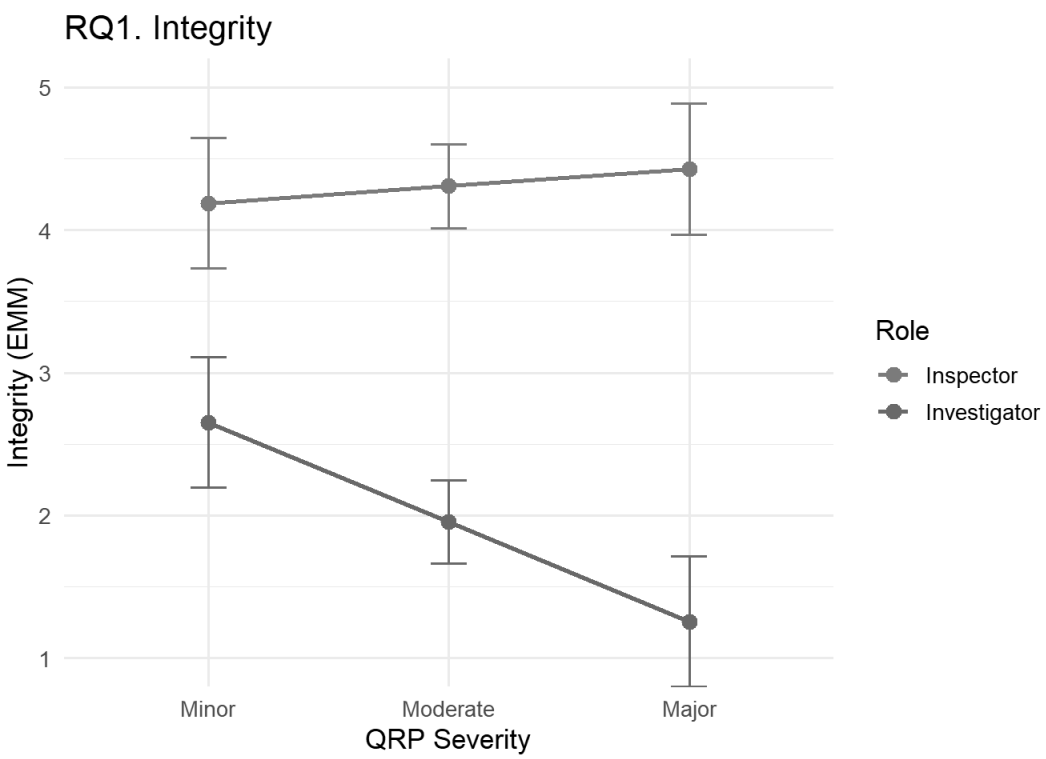
```
# Cohen's d for each role contrast.
eff_size(
  emm_Integrity,
  sigma = sigma(T4),
  edf = df.residual(T4)
)
```

```
## severity_num = -1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -2.51 0.503 16.8    -3.57    -1.45
##
## severity_num =  0:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -3.84 0.325 16.3    -4.53    -3.15
##
## severity_num =  1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -5.17 0.507 16.8    -6.24    -4.10
##
## sigma used for effect sizes: 0.6128
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

###figure

```
emm_Integrity_df = as.data.frame(
  emmeans(
    T4,
    ~ role_ec * severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Role = if_else(
      role_ec == 1,
      "Inspector",
      "Investigator"
    ),
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_Integrity_df,
  aes(
    x = Severity,
    y = emmean,
    color = Role,
    group = Role
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.15
  ) +
  scale_color_manual(
    values = c(
      "Inspector" = "#2196F3",
      "Investigator" = "#E53935"
    )
  ) +
  labs(
    title = "RQ1. Integrity",
    y = "Integrity (EMM)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 13) +
  coord_cartesian(ylim = c(1, 5))
```



##Likeability

Research Question 2: How is likeability influenced by role and QRP severity?

Hypothesis 2a: Investigators will be rated more likeable than inspectors.

Hypothesis 2b: At low QRP severity, investigators will be rated more likeable than inspectors, but at high QRP severity, inspectors will be rated more likeable than investigators.

###Random-effects structure for Likeability

```
# ------------------------------------------------------------
# Maximal participant-level structure
# ------------------------------------------------------------
L4_max1 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_max1_singular = isSingular(L4_max1, tol = 1e-4)
print(L4_max1_singular)
```

```
## [1] FALSE
```

```
# Step down: remove random interaction slope.
L4_max2 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec + severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
L4_max2_singular = isSingular(L4_max2, tol = 1e-4)
print(L4_max2_singular)
```

```
## [1] TRUE
```

```
# Step down: retain random role slope only.
L4_max3 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_max3_singular = isSingular(L4_max3, tol = 1e-4)
print(L4_max3_singular)
```

```
## [1] FALSE
```

```
# Test preregistered vignette random intercept with the maximal
# participant-level structure.
L4_prereg = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid) +
    (1 | vignette_id),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_prereg_singular = isSingular(L4_prereg, tol = 1e-4)
print(L4_prereg_singular)
```

```
## [1] FALSE
```

```
likeability_uses_vignette_term = !L4_prereg_singular

cat(
  "Likeability final random-effects structure includes vignette intercept:",
  likeability_uses_vignette_term,
  "\n"
)
```

```
## Likeability final random-effects structure includes vignette intercept: TRUE
```

###step-up models

```
# ------------------------------------------------------------
# Fit Models 0–4 using ONE identical random-effects structure.
# ------------------------------------------------------------

if (likeability_uses_vignette_term) {

  random_part_likeability = "participant_maximal_plus_vignette"

  M0_like = lmer(
    likeability ~ 1 +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L1 = lmer(
    likeability ~ role_ec +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L2 = lmer(
    likeability ~ severity_num +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L3 = lmer(
    likeability ~ role_ec + severity_num +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L4 = L4_prereg

} else {

  random_part_likeability = "participant_maximal"

  M0_like = lmer(
    likeability ~ 1 +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L1 = lmer(
    likeability ~ role_ec +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L2 = lmer(
    likeability ~ severity_num +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
```

```
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L3 = lmer(
    likeability ~ role_ec + severity_num +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L4 = L4_max1
}
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -8.1e+00
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -1.7e+01
```

```
cat("Likeability final model singularity:\n")
```

```
## Likeability final model singularity:
```

```
print(isSingular(M0_like))
```

```
## [1] TRUE
```

```
print(isSingular(L1))
```

```
## [1] FALSE
```

```
print(isSingular(L2))
```

```
## [1] FALSE
```

```
print(isSingular(L3))
```

```
## [1] TRUE
```

```
print(isSingular(L4))
```

```
## [1] FALSE
```

```
likeability_models = list(
  M0_like,
  L1,
  L2,
  L3,
  L4
)

likeability_model_names = c(
  "Model 0: Null",
  "Model 1: Role",
  "Model 2: QRP Severity",
  "Model 3: Role + QRP Severity",
  "Model 4: Role × QRP Severity"
)

likeability_fit = model_fit_table(
  likeability_models,
  likeability_model_names
)

print(likeability_fit)
```

```
##                          Model    AIC    BIC  logLik Chisq df      p
## 1                Model 0: Null 6191.7 6268.3 -3082.8     —  —      —
## 2                Model 1: Role 6183.5 6266.0 -3077.7 10.20  1  0.001
## 3        Model 2: QRP Severity 6193.5 6276.0 -3082.7  0.00  0      —
## 4 Model 3: Role + QRP Severity 6184.5 6272.9 -3077.2 10.99  1 < .001
## 5 Model 4: Role × QRP Severity 6169.5 6263.8 -3068.8 16.97  1 < .001
```

```
summary(L1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec + (1 + role_ec * severity_num | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6183.5    6266.0   -3077.7    6155.5      2664
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9403 -0.4843  0.0081  0.5067  4.1002
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.07706  0.27760
##              role_ec              0.24482  0.49480   0.26
##              severity_num         0.00554  0.07443  -0.53 -0.94
##              role_ec:severity_num 0.04783  0.21869  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.09784  0.31280
##  Residual                         0.35920  0.59933
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##             Estimate Std. Error     df t value Pr(>|t|)
## (Intercept)   3.1358     0.1289 6.0788  24.328 2.73e-07 ***
## role_ec       0.6663     0.1303 6.3558   5.112  0.00185 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##         (Intr)
## role_ec 0.005
```

```
summary(L2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## likeability ~ severity_num + (1 + role_ec * severity_num | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6193.5    6276.0   -3082.7    6165.5      2664
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9395 -0.4840  0.0108  0.5061  4.1041
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.07707  0.27761
##              role_ec              0.24487  0.49484   0.26
##              severity_num         0.00554  0.07443  -0.53 -0.94
##              role_ec:severity_num 0.04781  0.21865  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.52003  0.72113
##  Residual                         0.35920  0.59934
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1353     0.2949  5.9543  10.631 4.29e-05 ***
## severity_num  -0.1642     0.3609  5.9303  -0.455    0.665
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
summary(L3)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec + severity_num + (1 + role_ec * severity_num |
##     responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6184.5    6272.9   -3077.2    6154.5      2663
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8763 -0.4812  0.0096  0.5046  4.1504
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.077037 0.27756
##              role_ec              0.244774 0.49475   0.26
##              severity_num         0.005438 0.07374  -0.53 -0.96
##              role_ec:severity_num 0.047782 0.21859  -0.02  0.23 -0.19
##  vignette_id (Intercept)          0.079666 0.28225
##  Residual                         0.359370 0.59947
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1359     0.1166  6.0920  26.906 1.45e-07 ***
## role_ec        0.6666     0.1181  6.4325   5.642  0.00105 **
## severity_num  -0.1647     0.1419  5.9457  -1.161  0.29016
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c
## role_ec      0.006
## severity_nm -0.001 -0.005
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
summary(L4)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec * severity_num + (1 + role_ec * severity_num |
##     responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6169.5    6263.8   -3068.8    6137.5      2662
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9512 -0.4872  0.0110  0.5102  4.1226
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.076949 0.27740
##              role_ec              0.244494 0.49446   0.26
##              severity_num         0.005531 0.07437  -0.53 -0.94
##              role_ec:severity_num 0.047707 0.21842  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.004040 0.06356
##  Residual                         0.359310 0.59942
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           3.13589    0.03130  8.50866 100.189 2.26e-14 ***
## role_ec               0.66755    0.03680 16.13070  18.140 3.72e-12 ***
## severity_num         -0.16481    0.03498  5.91609  -4.711  0.00341 **
## role_ec:severity_num  0.33805    0.03631  6.85648   9.310 3.89e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c svrty_
## role_ec      0.070
## severity_nm -0.023 -0.060
## rl_c:svrty_ -0.003  0.041 -0.009
```

###fixed effects and planned role contrasts

```
likeability_fixed_effects = lapply(
  likeability_models,
  get_fixed_effects
)

names(likeability_fixed_effects) = paste0("Model_", 0:4)

print(likeability_fixed_effects)
```

```
## $Model_0
##     Parameter        b        SE       df        t            p
## 1 (Intercept) 3.135266 0.2999751 5.955134 10.45176 4.722539e-05
##
## $Model_1
##     Parameter         b        SE       df         t            p
## 1 (Intercept) 3.1357621 0.1288942 6.078846 24.328189 2.734277e-07
## 2     role_ec 0.6662713 0.1303398 6.355754  5.111804 1.849241e-03
##
## $Model_2
##      Parameter          b        SE       df          t            p
## 1  (Intercept)  3.1352764 0.2949213 5.954267 10.6308930 4.291348e-05
## 2 severity_num -0.1641912 0.3608626 5.930343 -0.4549963 6.652929e-01
##
## $Model_3
##      Parameter          b        SE       df         t            p
## 1  (Intercept)  3.1359306 0.1165518 6.091968 26.905897 1.451691e-07
## 2      role_ec  0.6666303 0.1181490 6.432463  5.642283 1.047393e-03
## 3 severity_num -0.1647117 0.1418814 5.945726 -1.160911 2.901565e-01
##
## $Model_4
##              Parameter          b         SE        df          t            p
## 1          (Intercept)  3.1358873 0.03129985  8.508664 100.188566 2.262092e-14
## 2              role_ec  0.6675478 0.03680049 16.130700  18.139642 3.720133e-12
## 3         severity_num -0.1648123 0.03498387  5.916089  -4.711094 3.412601e-03
## 4 role_ec:severity_num  0.3380490 0.03630913  6.856480   9.310302 3.891875e-05
```

```
# Planned role comparisons at each severity.
emm_like_role = emmeans(
  L4,
  pairwise ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_like_role$contrasts)
```

```
## severity_num = -1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -0.659 0.154 13.3  -4.275  0.0009
##
## severity_num =  0:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -1.335 0.104 12.6 -12.834 <0.0001
##
## severity_num =  1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -2.011 0.157 13.0 -12.814 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

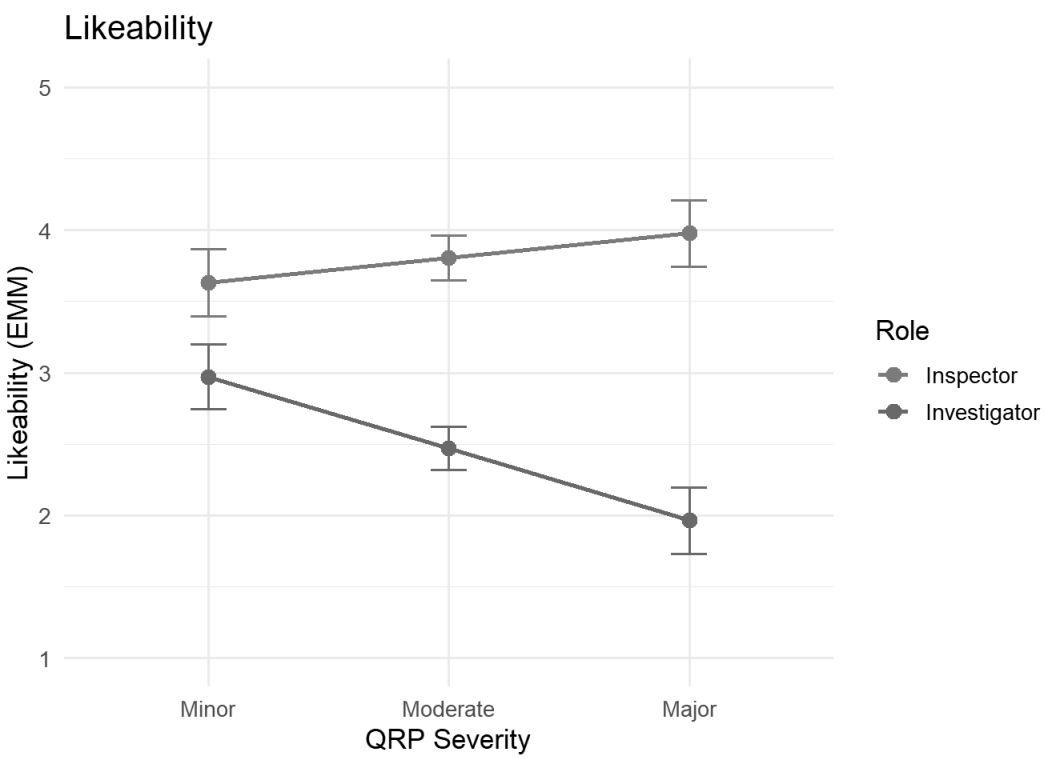
```
# Cohen's d for each role comparison.
eff_size(
  emmeans(
    L4,
    ~ role_ec | severity_num,
    at = list(severity_num = c(-1, 0, 1))
  ),
  sigma = sigma(L4),
  edf = df.residual(L4)
)
```

```
## severity_num = -1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -1.10 0.258 13.3    -1.65   -0.544
##
## severity_num =  0:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -2.23 0.176 12.7    -2.61   -1.846
##
## severity_num =  1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -3.36 0.266 13.3    -3.93   -2.782
##
## sigma used for effect sizes: 0.5994
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

###figure

```
emm_like_df = as.data.frame(
  emmeans(
    L4,
    ~ role_ec * severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Role = if_else(
      role_ec == 1,
      "Inspector",
      "Investigator"
    ),
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_like_df,
  aes(
    x = Severity,
    y = emmean,
    color = Role,
    group = Role
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.15
  ) +
  scale_color_manual(
    values = c(
      "Inspector" = "#2196F3",
      "Investigator" = "#E53935"
    )
  ) +
  labs(
    title = "Likeability",
    y = "Likeability (EMM)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 13) +
  coord_cartesian(ylim = c(1, 5))
```



##TRUENESS

Research Question 3: Is the perceived trueness of the finding influenced by QRP severity?**

Hypothesis 3: Perceived trueness will decrease as QRP severity increases.

###data and model

```r
# Trueness is only rated for investigators, with one item per vignette.
trueness_dat = filtered_by_age %>%
  transmute(
    responseid,
    minor = parse_num(kcon),
    moderate = parse_num(hcon),
    major = parse_num(xcon)
  ) %>%
  pivot_longer(
    cols = c(minor, moderate, major),
    names_to = "severity",
    values_to = "trueness"
  ) %>%
  mutate(
    severity = factor(
      severity,
      levels = c("minor", "moderate", "major")
    ),
    severity_num = case_when(
      severity == "minor" ~ -1,
      severity == "moderate" ~ 0,
      severity == "major" ~ 1
    ),
    responseid = factor(responseid),
    trueness_c = trueness - 3
  )


# Descriptives.
trueness_descriptives = trueness_dat %>%
  group_by(severity) %>%
  summarise(
    n = sum(!is.na(trueness)),
    M = round(mean(trueness, na.rm = TRUE), 2),
    SD = round(sd(trueness, na.rm = TRUE), 2),
    .groups = "drop"
  )

print(trueness_descriptives)
```

```
## # A tibble: 3 × 4
##   severity     n     M    SD
##   <fct>    <int> <dbl> <dbl>
## 1 minor      446  2.98  0.91
## 2 moderate   447  1.86  0.92
## 3 major      447  1.53  0.88
```

```r
# FT1: perceived trueness predicted by QRP severity.
FT1 = lmer(
  trueness_c ~ severity_num +
    (1 | responseid) +
    (1 | severity),
  data = trueness_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

summary(FT1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: trueness_c ~ severity_num + (1 | responseid) + (1 | severity)
##    Data: trueness_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    3503.7    3529.7   -1746.8    3493.7      1335
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7309 -0.6556 -0.0754  0.4459  4.0079
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  responseid (Intercept) 0.16689  0.4085
##  severity   (Intercept) 0.03313  0.1820
##  Residual               0.65229  0.8076
## Number of obs: 1340, groups:  responseid, 448; severity, 3
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.8771     0.1091  3.1960  -8.040  0.00318 **
## severity_num  -0.7213     0.1315  2.9993  -5.485  0.01193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
isSingular(FT1)
```

```
## [1] FALSE
```

```
# Fixed effects, including Satterthwaite df, t, and p.
trueness_fixed_effects = get_fixed_effects(FT1)
print(trueness_fixed_effects)
```

```
##      Parameter          b        SE       df         t           p
## 1  (Intercept) -0.8770718 0.1090937 3.195991 -8.039618 0.003179199
## 2 severity_num -0.7212843 0.1315047 2.999292 -5.484857 0.011928040
```

```
# Planned comparisons across severity levels.
emm_trueness = emmeans(
  FT1,
  pairwise ~ severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_trueness$contrasts)
```

```
##  contrast                        estimate    SE   df t.ratio p.value
##  (severity_num-1) - severity_num0    0.721 0.227 8.81   3.183  0.0343
##  (severity_num-1) - severity_num1    1.443 0.453 8.81   3.183  0.0343
##  severity_num0 - severity_num1       0.721 0.227 8.81   3.183  0.0343
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

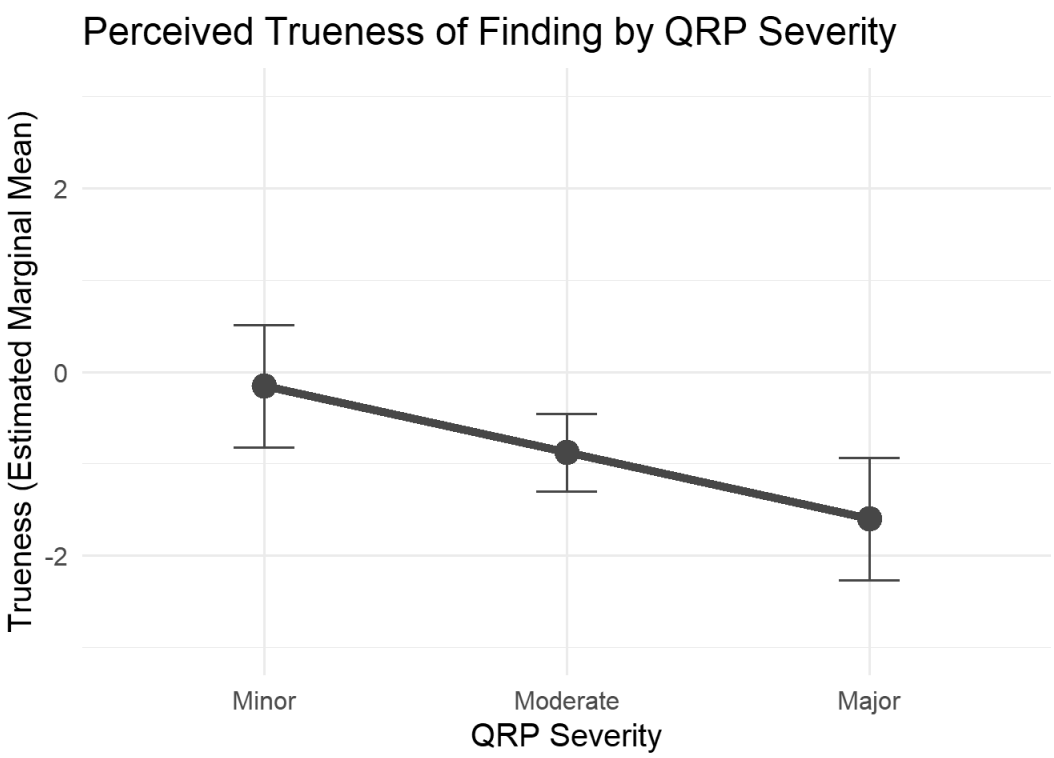
```
eff_size(
  emmeans(
    FT1,
    ~ severity_num,
    at = list(severity_num = c(-1, 0, 1))
  ),
  sigma = sigma(FT1),
  edf = df.residual(FT1)
)
```

```
##  contrast                        effect.size    SE   df lower.CL upper.CL
##  (severity_num-1) - severity_num0       0.893 0.281 8.64    0.253     1.53
##  (severity_num-1) - severity_num1       1.786 0.562 8.74    0.509     3.06
##  severity_num0 - severity_num1          0.893 0.281 8.64    0.253     1.53
##
## sigma used for effect sizes: 0.8076
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

###figure

```
emm_true_df = as.data.frame(
  emmeans(
    FT1,
    ~ severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_true_df,
  aes(
    x = Severity,
    y = emmean,
    group = 1
  )
) +
  geom_line(linewidth = 2, color = "#7B1FA2") +
  geom_point(size = 5, color = "#7B1FA2") +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.20,
    color = "#7B1FA2"
  ) +
  labs(
    title = "Perceived Trueness of Finding by QRP Severity",
    y = "Trueness (Estimated Marginal Mean)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 15) +
  coord_cartesian(ylim = c(-3, 3))
```



##Robustness checks — demographic covariates

```
demo_dat = filtered_by_age %>%
  transmute(
    responseid = responseid,
    gender = factor(`gender.identity`),
    political = factor(
      political,
      levels = c(
        "Strongly Liberal",
        "Liberal",
        "Lean Liberal",
        "Centrist",
        "Lean Conservative",
        "Conservative",
        "Strongly Conservative"
      )
    ),
    religion = factor(religion)
  )

tl_robust = tl_dat %>%
  mutate(responseid = as.character(responseid)) %>%
  left_join(demo_dat, by = "responseid") %>%
  mutate(responseid = factor(responseid))


# Integrity robustness model uses the exact random-effects structure

if (Integrity_uses_vignette_term) {

  T4_robust = lmer(
    Integrity ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

} else {

  T4_robust = lmer(
    Integrity ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec | responseid),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
}


# Likeability robustness model uses the exact random-effects structure

if (likeability_uses_vignette_term) {

  L4_robust = lmer(
    likeability ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

} else {

  L4_robust = lmer(
    likeability ~ role_ec * severity_num +
```

```
      gender + political + religion +
      (1 + role_ec * severity_num | responseid),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
}
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -3.2e-01
```

```
summary(T4_robust)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec * severity_num + gender + political + religion +
##     (1 + role_ec | responseid) + (1 | vignette_id)
##    Data: tl_robust
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5599.2    5733.8   -2776.6    5553.2      2551
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9084 -0.5233  0.0089  0.4967  5.4390
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.03990  0.1998
##              role_ec     0.16828  0.4102   0.33
##  vignette_id (Intercept) 0.01821  0.1350
##  Residual                0.37452  0.6120
## Number of obs: 2574, groups:  responseid, 429; vignette_id, 6
##
## Fixed effects:
##                                    Estimate Std. Error         df t value
## (Intercept)                         3.153051   0.101994  59.533220  30.914
## role_ec                             1.177933   0.059775   7.550335  19.706
## severity_num                       -0.292735   0.069073   5.986598  -4.238
## genderNon-binary / third gender    -0.277020   0.230884 428.135496  -1.200
## genderPrefer not to say            -0.042871   0.231147 428.135496  -0.185
## genderWoman                        -0.015940   0.040382 428.135496  -0.395
## politicalLiberal                   -0.046542   0.082963 428.135496  -0.561
## politicalLean Liberal              -0.099518   0.088391 428.135496  -1.126
## politicalCentrist                  -0.072534   0.079578 428.135497  -0.911
## politicalLean Conservative         -0.071195   0.082430 428.135497  -0.864
## politicalConservative              -0.024063   0.078867 428.135497  -0.305
## politicalStrongly Conservative     -0.006314   0.086966 428.135497  -0.073
## religionIslam                       0.309681   0.319419 428.135495   0.970
## religionJudaism                     0.046641   0.226141 428.135495   0.206
## religionNot Affiliated             -0.009012   0.057244 428.135496  -0.157
## religionOther                       0.135342   0.071001 428.135495   1.906
## religionProtestantism (Christian)   0.060335   0.039364 428.135496   1.533
## role_ec:severity_num                0.416278   0.069073   5.986598   6.027
##                                   Pr(>|t|)
## (Intercept)                        < 2e-16 ***
## role_ec                           9.12e-08 ***
## severity_num                       0.00548 **
## genderNon-binary / third gender    0.23087
## genderPrefer not to say            0.85295
## genderWoman                        0.69325
## politicalLiberal                   0.57510
## politicalLean Liberal              0.26085
## politicalCentrist                  0.36256
## politicalLean Conservative         0.38824
## politicalConservative              0.76043
## politicalStrongly Conservative     0.94216
## religionIslam                      0.33284
## religionJudaism                    0.83669
## religionNot Affiliated             0.87498
## religionOther                      0.05729 .
## religionProtestantism (Christian)  0.12608
## role_ec:severity_num               0.00095 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 18 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
summary(L4_robust)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## likeability ~ role_ec * severity_num + gender + political + religion +
##     (1 + role_ec * severity_num | responseid) + (1 | vignette_id)
##    Data: tl_robust
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5940.8    6116.4   -2940.4    5880.8      2544
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.7935 -0.4820  0.0202  0.5210  3.7346
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.071454 0.26731
##              role_ec              0.242165 0.49210   0.26
##              severity_num         0.005723 0.07565  -0.52 -0.96
##              role_ec:severity_num 0.045720 0.21382  -0.02  0.24 -0.21
##  vignette_id (Intercept)          0.003778 0.06147
##  Residual                         0.361073 0.60089
## Number of obs: 2574, groups:  responseid, 429; vignette_id, 6
##
## Fixed effects:
##                                   Estimate Std. Error        df t value
## (Intercept)                        3.04077    0.10041 325.78946  30.283
## role_ec                            0.66608    0.03653  17.13213  18.234
## severity_num                      -0.16725    0.03418   5.91030  -4.893
## genderNon-binary / third gender   -0.04049    0.26151 429.00555  -0.155
## genderPrefer not to say            0.29969    0.26181 429.00555   1.145
## genderWoman                        0.03465    0.04574 429.00556   0.757
## politicalLiberal                   0.05195    0.09397 429.00556   0.553
## politicalLean Liberal             -0.02167    0.10012 429.00556  -0.216
## politicalCentrist                 -0.03189    0.09014 429.00556  -0.354
## politicalLean Conservative        -0.01950    0.09337 429.00556  -0.209
## politicalConservative              0.02450    0.08933 429.00556   0.274
## politicalStrongly Conservative     0.06482    0.09850 429.00556   0.658
## religionIslam                      0.51745    0.36179 429.00555   1.430
## religionJudaism                    0.14722    0.25614 429.00555   0.575
## religionNot Affiliated            -0.03877    0.06484 429.00556  -0.598
## religionOther                      0.17098    0.08042 429.00555   2.126
## religionProtestantism (Christian)  0.08186    0.04459 429.00556   1.836
## role_ec:severity_num               0.34557    0.03552   6.88122   9.730
##                                  Pr(>|t|)
## (Intercept)                       < 2e-16 ***
## role_ec                          1.17e-12 ***
## severity_num                      0.00285 **
## genderNon-binary / third gender   0.87702
## genderPrefer not to say           0.25297
## genderWoman                       0.44917
## politicalLiberal                  0.58069
## politicalLean Liberal             0.82875
## politicalCentrist                 0.72369
## politicalLean Conservative        0.83470
## politicalConservative             0.78400
## politicalStrongly Conservative    0.51084
## religionIslam                     0.15338
## religionJudaism                   0.56575
## religionNot Affiliated            0.55019
## religionOther                     0.03407 *
## religionProtestantism (Christian) 0.06704 .
## role_ec:severity_num              2.86e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 18 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# Compare interaction coefficients with and without demographic covariates.
cat("Integrity interaction, original:",
    fixef(T4)[["role_ec:severity_num"]], "\n")
```

```
## Integrity interaction, original: 0.4080529
```

```
cat("Integrity interaction, adjusted:",
    fixef(T4_robust)[["role_ec:severity_num"]], "\n")
```

```
## Integrity interaction, adjusted: 0.4162782
```

```
cat("Likeability interaction, original:",
    fixef(L4)[["role_ec:severity_num"]], "\n")
```

```
## Likeability interaction, original: 0.338049
```

```
cat("Likeability interaction, adjusted:",
    fixef(L4_robust)[["role_ec:severity_num"]], "\n")
```

```
## Likeability interaction, adjusted: 0.3455711
```

APA TABLES

```r
#APA Model Comparison Tables

# Helper: extract fixed effects from a model into a clean data frame
fe_table = function(model) {
  cf = summary(model)$coefficients
  ci = confint(model, method = "Wald")
  ci = ci[rownames(cf), ]

  data.frame(
    Parameter = rownames(cf),
    b         = round(cf[, "Estimate"], 2),
    SE        = round(cf[, "Std. Error"], 2),
    df        = round(cf[, "df"], 2),
    CI_low    = round(ci[, 1], 2),
    CI_high   = round(ci[, 2], 2),
    t         = round(cf[, "t value"], 2),
    p         = ifelse(
      cf[, "Pr(>|t|)"] < .001,
      "< .001",
      formatC(cf[, "Pr(>|t|)"], digits = 3, format = "f")
    )
  )
}


relabel = function(df) {
  df$Parameter = dplyr::recode(
    df$Parameter,
    "(Intercept)"          = "Intercept",
    "role_ec"              = "Role",
    "severity_num"         = "QRP Severity",
    "role_ec:severity_num" = "Role × QRP Severity"
  )
  df
}


fit_table = function(model_list, model_names) {

  do.call(
    rbind,
    lapply(seq_along(model_list), function(i) {

      m = model_list[[i]]

      if (i == 1) {

        chisq_out = NA_character_
        df_out    = NA_character_
        p_out     = NA_character_

      } else {

        lr = anova(
          model_list[[i - 1]],
          m
        )

        chisq_out = formatC(
          lr$Chisq[2],
          digits = 2,
          format = "f"
        )

        df_out = as.character(
          lr$Df[2]
        )

        p_out = ifelse(
          lr$`Pr(>Chisq)`[2] < .001,
```

```
          "< .001",
          formatC(
            lr$`Pr(>Chisq)`[2],
            digits = 3,
            format = "f"
          )
        )
      }

      data.frame(
        Model  = model_names[i],
        AIC    = round(AIC(m), 1),
        BIC    = round(BIC(m), 1),
        logLik = round(as.numeric(logLik(m)), 1),
        Chisq  = chisq_out,
        df     = df_out,
        p      = p_out
      )
    })
  )
}


#TABLE 1: Integrity model fit

fit_table(
  list(
    T1,
    T2,
    T3,
    T4
  ),
  c(
    "M1: Role",
    "M2: Severity",
    "M3: Role + Severity",
    "M4: Role × Severity"
  )
) %>%

  kable(
    caption = "Table 1. Model Fit Indices for Step-Up Integrity Models",
    col.names = c(
      "Model",
      "AIC",
      "BIC",
      "logLik",
      "$\\chi^2$",
      "df",
      "p"
    ),
    align = "lrrrrrr",
    escape = FALSE
  ) %>%

  kable_styling(
    full_width = FALSE,
    bootstrap_options = "condensed"
  ) %>%

  footnote(
    general =
      "Lower AIC/BIC indicates better model fit. Chi-square tests reflect likelihood ratio comparison to the preceding mode
l."
  )
```

Table 1. Model Fit Indices for Step-Up Integrity Models

Model	AIC	BIC	logLik	χ^2	df	p
M1: Role	5815.5	5856.7	-2900.7	NA	NA	NA
M2: Severity	5828.1	5869.3	-2907.0	0.00	0	NA
M3: Role + Severity	5815.3	5862.5	-2899.7	14.72	1	< .001
M4: Role × Severity	5805.8	5858.9	-2893.9	11.52	1	< .001

Note:
Lower AIC/BIC indicates better model fit. Chi-square tests reflect likelihood ratio comparison to the preceding model.

```
#TABLE 2: Integrity fixed effects (Model 4)

fe_table(T4) %>%

  relabel() %>%

  mutate(
    `95% CI` =
      paste0(
        "[",
        CI_low,
        ", ",
        CI_high,
        "]"
      )
  ) %>%

  select(
    Parameter,
    b,
    SE,
    df,
    `95% CI`,
    t,
    p
  ) %>%

  kable(
    caption =
      "Table 2. Fixed Effects for Integrity Model 4 (Role × QRP Severity)",

    col.names = c(
      "Parameter",
      "b",
      "SE",
      "df",
      "95% CI",
      "t",
      "p"
    ),

    align = "lrrrrrr"
  ) %>%

  kable_styling(
    full_width = FALSE,
    bootstrap_options = "condensed"
  ) %>%

  footnote(
    general =
      "Role: inspector = 1, investigator = −1. QRP Severity: minor = −1, moderate = 0, major = 1."
  )
```

Table 2. Fixed Effects for Integrity Model 4 (Role × QRP Severity)

	Parameter	b	SE	df	95% CI	t	p
(Intercept)	Intercept	3.13	0.06	6.34	[3.02, 3.24]	54.79	< .001
role_ec	Role	1.18	0.06	7.48	[1.06, 1.29]	19.75	< .001
severity_num	QRP Severity	-0.29	0.07	5.99	[-0.42, -0.15]	-4.19	0.006
role_ec:severity_num	Role × QRP Severity	0.41	0.07	5.99	[0.27, 0.54]	5.92	0.001

Note:

Role: inspector = 1, investigator = −1. QRP Severity: minor = −1, moderate = 0, major = 1.

```
# TABLE 3: Likeability model fit

fit_table(
  list(
    L1,
    L2,
    L3,
    L4
  ),
  c(
    "M1: Role",
    "M2: Severity",
    "M3: Role + Severity",
    "M4: Role × Severity"
  )
) %>%

  kable(
    caption =
      "Table 3. Model Fit Indices for Step-Up Likeability Models",

    col.names = c(
      "Model",
      "AIC",
      "BIC",
      "logLik",
      "$\\chi^2$",
      "df",
      "p"
    ),

    align = "lrrrrrr",
    escape = FALSE
  ) %>%

  kable_styling(
    full_width = FALSE,
    bootstrap_options = "condensed"
  ) %>%

  footnote(
    general =
      "Lower AIC/BIC indicates better model fit. Chi-square tests reflect likelihood ratio comparison to the preceding mode
l."
  )
```

Table 3. Model Fit Indices for Step-Up Likeability Models

Model	AIC	BIC	logLik	χ^2	df	p
M1: Role	6183.5	6266.0	-3077.7	NA	NA	NA
M2: Severity	6193.5	6276.0	-3082.7	0.00	0	NA
M3: Role + Severity	6184.5	6272.9	-3077.2	10.99	1	< .001
M4: Role × Severity	6169.5	6263.8	-3068.8	16.97	1	< .001

Note:

Lower AIC/BIC indicates better model fit. Chi-square tests reflect likelihood ratio comparison to the preceding model.

```
#TABLE 4: Likeability fixed effects (Model 4)

fe_table(L4) %>%

  relabel() %>%

  mutate(
    `95% CI` =
      paste0(
        "[",
        CI_low,
        ", ",
        CI_high,
        "]"
      )
  ) %>%

  select(
    Parameter,
    b,
    SE,
    df,
    `95% CI`,
    t,
    p
  ) %>%

  kable(
    caption =
      "Table 4. Fixed Effects for Likeability Model 4 (Role × QRP Severity)",

    col.names = c(
      "Parameter",
      "b",
      "SE",
      "df",
      "95% CI",
      "t",
      "p"
    ),

    align = "lrrrrrr"
  ) %>%

  kable_styling(
    full_width = FALSE,
    bootstrap_options = "condensed"
  ) %>%

  footnote(
    general =
      "Role: inspector = 1, investigator = −1. QRP Severity: minor = −1, moderate = 0, major = 1."
  )
```

Table 4. Fixed Effects for Likeability Model 4 (Role × QRP Severity)

	Parameter	b	SE	df	95% CI	t	p
(Intercept)	Intercept	3.14	0.03	8.51	[3.07, 3.2]	100.19	< .001
role_ec	Role	0.67	0.04	16.13	[0.6, 0.74]	18.14	< .001
severity_num	QRP Severity	-0.16	0.03	5.92	[-0.23, -0.1]	-4.71	0.003
role_ec:severity_num	Role × QRP Severity	0.34	0.04	6.86	[0.27, 0.41]	9.31	< .001

Note:

Role: inspector = 1, investigator = −1. QRP Severity: minor = −1, moderate = 0, major = 1.

```
#TABLE 5: Trueness fixed effects (FT1)

fe_table(FT1) %>%

  relabel() %>%

  mutate(
    `95% CI` =
      paste0(
        "[",
        CI_low,
        ", ",
        CI_high,
        "]"
      )
  ) %>%

  select(
    Parameter,
    b,
    SE,
    df,
    `95% CI`,
    t,
    p
  ) %>%

  kable(
    caption =
      "Table 5. Fixed Effects for Trueness Model FT1",

    col.names = c(
      "Parameter",
      "b",
      "SE",
      "df",
      "95% CI",
      "t",
      "p"
    ),

    align = "lrrrrrr"
  ) %>%

  kable_styling(
    full_width = FALSE,
    bootstrap_options = "condensed"
  ) %>%

  footnote(
    general =
      "Trueness centered at 3 (neutral midpoint). QRP Severity: minor = −1, moderate = 0, major = 1."
  )
```

Table 5. Fixed Effects for Trueness Model FT1

	Parameter	b	SE	df	95% CI	t	p
(Intercept)	Intercept	-0.88	0.11	3.2	[-1.09, -0.66]	-8.04	0.003
severity_num	QRP Severity	-0.72	0.13	3.0	[-0.98, -0.46]	-5.48	0.012

Note:

Trueness centered at 3 (neutral midpoint). QRP Severity: minor = −1, moderate = 0, major = 1.

```r
get_fe = function(model) {

  cf = summary(model)$coefficients

  list(
    b  = setNames(
      round(cf[, "Estimate"], 2),
      rownames(cf)
    ),

    se = setNames(
      round(cf[, "Std. Error"], 2),
      rownames(cf)
    ),

    df = if ("df" %in% colnames(cf)) {
      setNames(
        round(cf[, "df"], 2),
        rownames(cf)
      )
    } else {
      NULL
    },

    t = if ("t value" %in% colnames(cf)) {
      setNames(
        round(cf[, "t value"], 2),
        rownames(cf)
      )
    } else {
      NULL
    },

    p = setNames(
      cf[, "Pr(>|t|)"],
      rownames(cf)
    )
  )
}

#format p value

fmt_p = function(p) {

  if (is.na(p)) return("")

  if (p < .001) {
    return("< .001***")
  }

  stars =
    ifelse(
      p < .01, "**",
      ifelse(
        p < .05, "*",
        ""
      )
    )

  paste0(
    formatC(
      p,
      digits = 3,
      format = "f"
    ),
    stars
  )
}
```

```r
#extract one fixed effect

cell = function(fe, term) {

  if (!term %in% names(fe$b)) {
    return(
      c("", "", "")
    )
  }

  c(
    as.character(fe$b[term]),
    as.character(fe$se[term]),
    fmt_p(fe$p[term])
  )
}

#Random effects

get_re = function(model) {

  vc = as.data.frame(
    lme4::VarCorr(model)
  )

  setNames(
    round(vc$vcov, 3),
    vc$grp
  )
}


#ICC

get_icc = function(model) {

  vc = as.data.frame(
    lme4::VarCorr(model)
  )

  total = sum(vc$vcov)

  setNames(
    round(
      vc$vcov / total,
      3
    ),
    vc$grp
  )
}

#table set up

make_stepup_table = function(
    m0,
    m1,
    m2,
    m3,
    m4,
    caption_text,
    footnote_text
) {

  models = list(
    m0,
    m1,
    m2,
    m3,
    m4
  )
```

```
  fe_list = lapply(
    models,
    get_fe
  )

  re_list = lapply(
    models,
    get_re
  )

  icc_list = lapply(
    models,
    get_icc
  )

#fixed effect

  term_labels = c(
    "(Intercept)" =
      "Intercept",

    "role_ec" =
      "Role",

    "severity_num" =
      "QRP Severity",

    "role_ec:severity_num" =
      "Role × QRP Severity"
  )


  rows = lapply(
    names(term_labels),
    function(term) {

      cells =
        do.call(
          cbind,
          lapply(
            fe_list,
            function(fe) cell(fe, term)
          )
        )

      data.frame(

        Parameter =
          term_labels[term],

        M0_b =
          cells[1, 1],

        M0_se =
          cells[2, 1],

        M0_p =
          cells[3, 1],

        M1_b =
          cells[1, 2],

        M1_se =
          cells[2, 2],

        M1_p =
          cells[3, 2],

        M2_b =
          cells[1, 3],
```

```
      M2_se =
        cells[2, 3],

      M2_p =
        cells[3, 3],

      M3_b =
        cells[1, 4],

      M3_se =
        cells[2, 4],

      M3_p =
        cells[3, 4],

      M4_b =
        cells[1, 5],

      M4_se =
        cells[2, 5],

      M4_p =
        cells[3, 5],

      stringsAsFactors = FALSE
    )
  }
)


fe_df =
  do.call(
    rbind,
    rows
  )


#random effects

re_grps = c(

  "responseid" =
    "Participant variance",

  "vignette_id" =
    "Vignette variance",

  "Residual" =
    "Residual variance"
)


re_rows =
  lapply(
    names(re_grps),
    function(grp) {

      vals =
        sapply(
          re_list,
          function(re) {

            if (grp %in% names(re)) {

              as.character(
                re[grp]
              )

            } else {

              "–"
```

```
            }
          }
        )

      data.frame(

        Parameter =
          re_grps[grp],

        M0_b = vals[1],
        M0_se = "",
        M0_p = "",

        M1_b = vals[2],
        M1_se = "",
        M1_p = "",

        M2_b = vals[3],
        M2_se = "",
        M2_p = "",

        M3_b = vals[4],
        M3_se = "",
        M3_p = "",

        M4_b = vals[5],
        M4_se = "",
        M4_p = "",

        stringsAsFactors = FALSE
      )
    }
  )


re_df =
  do.call(
    rbind,
    re_rows
  )


#ICC

icc_rows =
  lapply(
    c(
      "responseid",
      "vignette_id"
    ),
    function(grp) {

      vals =
        sapply(
          icc_list,
          function(ic) {

            if (grp %in% names(ic)) {

              as.character(
                ic[grp]
              )

            } else {

              "-"
            }
          }
        )

      label =
```

```
          ifelse(
            grp == "responseid",
            "ICC: Participant",
            "ICC: Vignette"
          )

        data.frame(

          Parameter = label,

          M0_b = vals[1],
          M0_se = "",
          M0_p = "",

          M1_b = vals[2],
          M1_se = "",
          M1_p = "",

          M2_b = vals[3],
          M2_se = "",
          M2_p = "",

          M3_b = vals[4],
          M3_se = "",
          M3_p = "",

          M4_b = vals[5],
          M4_se = "",
          M4_p = "",

          stringsAsFactors = FALSE
        )
      }
    )


  icc_df =
    do.call(
      rbind,
      icc_rows
    )


# AIC and BIC
  fit_vals =
    sapply(
      models,
      function(m) {

        c(
          AIC = round(
            AIC(m),
            1
          ),

          BIC = round(
            BIC(m),
            1
          )
        )
      }
    )


  fit_df =
    data.frame(

      Parameter =
        c(
          "AIC",
          "BIC"
```

```
      ),

    M0_b =
      as.character(
        fit_vals[, 1]
      ),

    M0_se = "",
    M0_p = "",

    M1_b =
      as.character(
        fit_vals[, 2]
      ),

    M1_se = "",
    M1_p = "",

    M2_b =
      as.character(
        fit_vals[, 3]
      ),

    M2_se = "",
    M2_p = "",

    M3_b =
      as.character(
        fit_vals[, 4]
      ),

    M3_se = "",
    M3_p = "",

    M4_b =
      as.character(
        fit_vals[, 5]
      ),

    M4_se = "",
    M4_p = "",

    stringsAsFactors = FALSE
  )

#model comparison

  comparisons = list()

  for (i in 2:5) {

    comp =
      anova(
        models[[i - 1]],
        models[[i]]
      )

    chi =
      comp$Chisq[2]

    df =
      comp$Df[2]

    p =
      comp$`Pr(>Chisq)`[2]

    comparisons[[i - 1]] =
      c(
        chi,
        df,
        p
```

```
    )
}


comparison_df =
  data.frame(

    Parameter =
      c(
        "Model comparison χ²",
        "Model comparison df",
        "Model comparison p"
      ),

    M0_b = "—",
    M0_se = "",
    M0_p = "",

    M1_b =
      as.character(
        round(
          comparisons[[1]][1],
          2
        )
      ),

    M1_se =
      as.character(
        comparisons[[1]][2]
      ),

    M1_p =
      fmt_p(
        comparisons[[1]][3]
      ),

    M2_b =
      as.character(
        round(
          comparisons[[2]][1],
          2
        )
      ),

    M2_se =
      as.character(
        comparisons[[2]][2]
      ),

    M2_p =
      fmt_p(
        comparisons[[2]][3]
      ),

    M3_b =
      as.character(
        round(
          comparisons[[3]][1],
          2
        )
      ),

    M3_se =
      as.character(
        comparisons[[3]][2]
      ),

    M3_p =
      fmt_p(
        comparisons[[3]][3]
      ),
```

```
    M4_b =
      as.character(
        round(
          comparisons[[4]][1],
          2
        )
      ),

    M4_se =
      as.character(
        comparisons[[4]][2]
      ),

    M4_p =
      fmt_p(
        comparisons[[4]][3]
      ),

    stringsAsFactors = FALSE
  )


full_df =
  rbind(
    fe_df,
    re_df,
    icc_df,
    fit_df,
    comparison_df
  )


full_df %>%

  kable(

    caption =
      caption_text,

    col.names =
      c(
        "Parameter",

        "b", "SE", "p",
        "b", "SE", "p",
        "b", "SE", "p",
        "b", "SE", "p",
        "b", "SE", "p"
      ),

    align =
      c(
        "l",
        rep(
          "r",
          15
        )
      ),

    escape = FALSE
  ) %>%

  kable_styling(

    full_width = TRUE,

    font_size = 11,

    bootstrap_options =
      c(
```

```
          "condensed",
          "bordered"
        )
    ) %>%

add_header_above(
  c(
    " " = 2,
    "Model 0" = 3,
    "Model 1" = 3,
    "Model 2" = 3,
    "Model 3" = 3,
    "Model 4" = 3
  )
    ) %>%

    footnote(

      general =
        footnote_text,

      general_title =
        "Note.",

      footnote_as_chunk =
        TRUE
    )
}
```

```
make_stepup_table(

  m0 = M0_Integrity,
  m1 = T1,
  m2 = T2,
  m3 = T3,
  m4 = T4,

  caption_text =
    "Table 8. Step-Up MLM Results for Integrity Ratings",

  footnote_text =
    "Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate =
0, and major = 1. Model comparison χ² values are likelihood-ratio tests comparing each model with the preceding model. * p <
.05. ** p < .01. *** p < .001."
)
```

Table 8. Step-Up MLM Results for Integrity Ratings

		Model 0			Model 1			Model 2			Model 3			Model 4		
	Parameter	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**
(Intercept)	Intercept	3.13	0.51	< .001***	3.13	0.18	< .001***	3.13	0.5	< .001***	3.13	0.15	< .001***	3.13	0.06	< .001***
role_ec	Role				1.18	0.18	< .001***				1.18	0.15	< .001***	1.18	0.06	< .001***
severity_num	QRP Severity							-0.29	0.61	0.655	-0.29	0.18	0.160	-0.29	0.07	0.006**
role_ec:severity_num	Role × QRP Severity													0.41	0.07	0.001**
responseid	Participant variance	0.041			0.041			0.041			0.041			0.041		
vignette_id	Vignette variance	1.565			0.185			1.509			0.129			0.018		
Residual	Residual variance	0.375			0.375			0.375			0.375			0.376		
1	ICC: Participant	0.019			0.052			0.019			0.055			0.065		

Note. Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. Model comparison χ² values are likelihood-ratio tests comparing each model with the preceding model. * p < .05. ** p < .01. *** p < .001.

		Model 0			Model 1			Model 2			Model 3			Model 4		
	Parameter	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**
2	ICC: Vignette	0.719			0.232			0.712			0.175			0.029		
11	AIC	5826.3			5815.5			5828.1			5815.3			5805.8		
21	BIC	5861.6			5856.7			5869.3			5862.5			5858.9		
12	Model comparison χ^2	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***
22	Model comparison df	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***
3	Model comparison p	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***

Note. Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. Model comparison χ^2 values are likelihood-ratio tests comparing each model with the preceding model. * p < .05. ** p < .01. *** p < .001.

```
#LIKEABILITY TABLE

make_stepup_table(

  m0 = M0_like,
  m1 = L1,
  m2 = L2,
  m3 = L3,
  m4 = L4,

  caption_text =
    "Table 9. Step-Up MLM Results for Likeability Ratings",

  footnote_text =
    "Role was effect coded as inspector = 1 and investigator = -1. QRP Severity was effect coded as minor = -1, moderate =
0, and major = 1. Model comparison χ² values are likelihood-ratio tests comparing each model with the preceding model. * p <
.05. ** p < .01. *** p < .001."
)
```

Table 9. Step-Up MLM Results for Likeability Ratings

		Model 0			Model 1			Model 2			Model 3			Model 4		
	Parameter	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**
(Intercept)	Intercept	3.14	0.3	< .001***	3.14	0.13	< .001***	3.14	0.29	< .001***	3.14	0.12	< .001***	3.14	0.03	< .001***
role_ec	Role				0.67	0.13	0.002**				0.67	0.12	0.001**	0.67	0.04	< .001***
severity_num	QRP Severity							-0.16	0.36	0.665	-0.16	0.14	0.290	-0.16	0.03	0.003**
role_ec:severity_num	Role × QRP Severity													0.34	0.04	< .001***
responseid	Participant variance	0.077			0.077			0.077			0.077			0.077		
vignette_id	Vignette variance	0.538			0.098			0.52			0.08			0.004		
Residual	Residual variance	0.359			0.359			0.359			0.359			0.359		
1	ICC: Participant	0.06			0.092			0.061			0.093			0.103		
2	ICC: Vignette	0.42			0.117			0.412			0.097			0.005		
11	AIC	6191.7			6183.5			6193.5			6184.5			6169.5		
21	BIC	6268.3			6266			6276			6272.9			6263.8		
12	Model comparison χ^2	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***
22	Model comparison df	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***
3	Model comparison p	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***

Note. Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. Model comparison χ^2 values are likelihood-ratio tests comparing each model with the preceding model. * p < .05. ** p < .01. *** p < .001.

```r
# Fixed effects
ft1 <- summary(FT1)$coefficients

# Random effects
ft1_vc <- as.data.frame(
  lme4::VarCorr(FT1)
)

# Extract variance components
participant_var <- ft1_vc$vcov[
  ft1_vc$grp == "responseid"
]

vignette_var <- ft1_vc$vcov[
  ft1_vc$grp == "severity"
]

residual_var <- ft1_vc$vcov[
  ft1_vc$grp == "Residual"
]

# Total variance
total_var <-
  participant_var +
  vignette_var +
  residual_var

# ICCs
participant_icc <-
  participant_var / total_var

vignette_icc <-
  vignette_var / total_var


# Create table

trueness_df <- data.frame(

  Parameter = c(
    "Intercept",
    "QRP Severity",
    "Participant variance",
    "Vignette variance",
    "Residual variance",
    "ICC: Participant",
    "ICC: Vignette",
    "AIC",
    "BIC"
  ),

  b = c(
    sprintf(
      "%.2f",
      ft1["(Intercept)", "Estimate"]
    ),

    sprintf(
      "%.2f",
      ft1["severity_num", "Estimate"]
    ),

    sprintf(
      "%.3f",
      participant_var
    ),

    sprintf(
      "%.3f",
      vignette_var
```

```
    ),

    sprintf(
      "%.3f",
      residual_var
    ),

    sprintf(
      "%.3f",
      participant_icc
    ),

    sprintf(
      "%.3f",
      vignette_icc
    ),

    sprintf(
      "%.1f",
      AIC(FT1)
    ),

    sprintf(
      "%.1f",
      BIC(FT1)
    )
  ),

  SE = c(
    sprintf(
      "%.2f",
      ft1["(Intercept)", "Std. Error"]
    ),

    sprintf(
      "%.2f",
      ft1["severity_num", "Std. Error"]
    ),

    "", "", "", "", "", "", ""
  ),

  p = c(
    fmt_p(
      ft1["(Intercept)", "Pr(>|t|)"]
    ),

    fmt_p(
      ft1["severity_num", "Pr(>|t|)"]
    ),

    "", "", "", "", "", "", ""
  ),

  stringsAsFactors = FALSE
)

# Print Table 10

trueness_df %>%

  kable(
    caption =
      "Table 10. MLM Results for Perceived Trueness of Finding",

    col.names = c(
      "Parameter",
      "b",
      "SE",
      "p"
    ),
```

```
    align = c(
      "l",
      "r",
      "r",
      "r"
    ),

    escape = FALSE
  ) %>%

  kable_styling(
    full_width = FALSE,
    font_size = 11,
    bootstrap_options =
      c(
        "condensed",
        "bordered"
      )
  ) %>%

  footnote(
    general =
      "Trueness was centered at 3 (neutral). QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. * p <
.05. ** p < .01. *** p < .001.",

    general_title =
      "Note.",

    footnote_as_chunk =
      TRUE
  )
```

Table 10. MLM Results for Perceived Trueness of Finding

Parameter	b	SE	p
Intercept	-0.88	0.11	0.003**
QRP Severity	-0.72	0.13	0.012*
Participant variance	0.167		
Vignette variance	0.033		
Residual variance	0.652		
ICC: Participant	0.196		
ICC: Vignette	0.039		
AIC	3503.7		
BIC	3529.7		

Note. Trueness was centered at 3 (neutral). QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. * p < .05. ** p < .01. *** p < .001.

```
cat("\n--- Integrity random effects ---\n")
```

```
##
## --- Integrity random effects ---
```

```
print(get_variance_components(T4))
```

```
##          grp Variance    SD
## 1  responseid    0.041 0.202
## 2  responseid    0.168 0.410
## 3  responseid    0.026 0.316
## 4 vignette_id    0.018 0.135
## 5    Residual    0.376 0.613
```

```
print(get_random_intercept_icc(T4))
```

```
##          grp   ICC
## 1  responseid 0.049
## 2  responseid 0.031
## 3 vignette_id 0.022
## 4    Residual 0.449
```

```
cat("\n--- Likeability random effects ---\n")
```

```
##
## --- Likeability random effects ---
```

```
print(get_variance_components(L4))
```

```
##            grp Variance     SD
## 1   responseid    0.077  0.277
## 2   responseid    0.244  0.494
## 3   responseid    0.006  0.074
## 4   responseid    0.048  0.218
## 5   responseid    0.036  0.260
## 6   responseid   -0.011 -0.532
## 7   responseid   -0.002 -0.025
## 8   responseid   -0.035 -0.943
## 9   responseid    0.024  0.226
## 10  responseid   -0.005 -0.331
## 11 vignette_id    0.004  0.064
## 12    Residual    0.359  0.599
```

```
print(get_random_intercept_icc(L4))
```

```
##           grp    ICC
## 1  responseid  0.094
## 2  responseid  0.043
## 3  responseid -0.013
## 4  responseid -0.002
## 5 vignette_id  0.005
## 6    Residual  0.437
```

```
cat("\n--- Trueness random effects ---\n")
```

```
##
## --- Trueness random effects ---
```

```
print(get_variance_components(FT1))
```

```
##          grp Variance    SD
## 1 responseid    0.167 0.409
## 2   severity    0.033 0.182
## 3   Residual    0.652 0.808
```

```
print(get_random_intercept_icc(FT1))
```

```
##          grp   ICC
## 1 responseid 0.111
## 2   severity 0.022
## 3   Residual 0.434
```

#CFA

```r
# Load packages
library(dplyr)
library(tidyr)
library(lavaan)
```

```
## Warning: package 'lavaan' was built under R version 4.5.3
```

```
## This is lavaan 0.7-2
## lavaan is FREE software! Please report any bugs.
```

```
##
## Attaching package: 'lavaan'
```

```
## The following object is masked from 'package:psych':
##
##     cor2cov
```

```r
library(semTools)
```

```
## Warning: package 'semTools' was built under R version 4.5.3
```

```
##
```

```
## ###############################################################################
```

```
## This is semTools 0.5-9
```

```
## All users of R (or SEM) are invited to submit functions or ideas for functions.
```

```
## ###############################################################################
```

```
##
## Attaching package: 'semTools'
```

```
## The following objects are masked from 'package:psych':
##
##     reliability, skew
```

```
## The following object is masked from 'package:readr':
##
##     clipboard
```

```
# 1. Create participant-level data


cfa_dat <- tl_dat %>%
  group_by(responseid, role) %>%
  summarise(
    hon  = mean(hon_raw,  na.rm = TRUE),
    sin  = mean(sin_raw,  na.rm = TRUE),
    eth  = mean(eth_raw,  na.rm = TRUE),
    like = mean(like_raw, na.rm = TRUE),
    app  = mean(app_raw,  na.rm = TRUE),
    .groups = "drop"
  )


# 2. One-factor model

one_factor <- '
  Evaluation =~ hon + sin + eth + like + app
'



# Investigator: one-factor model
investigator_one <- cfa(
  one_factor,
  data = cfa_dat %>% filter(role == "investigator"),
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 57 108 184 279 383 420 431.
```

```
# Inspector: one-factor model
inspector_one <- cfa(
  one_factor,
  data = cfa_dat %>% filter(role == "inspector"),
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 57 108 172 184 279 383 420 431.
```

```
# 3. Two-factor model

two_factor <- '
  Integrity =~ hon + sin + eth
  Likeability =~ like + app
'



# Investigator: two-factor model
investigator_two <- cfa(
  two_factor,
  data = cfa_dat %>% filter(role == "investigator"),
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 57 108 184 279 383 420 431.
```

```
# Inspector: two-factor model
inspector_two <- cfa(
  two_factor,
  data = cfa_dat %>% filter(role == "inspector"),
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 57 108 172 184 279 383 420 431.
```

```
# 4. Print model results


cat("\n================ INVESTIGATOR: ONE FACTOR ================\n")
```

```
##
## ================ INVESTIGATOR: ONE FACTOR ================
```

```
summary(
  investigator_one,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 43 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        15
##
##                                                   Used       Total
##   Number of observations                           448         455
##   Number of missing patterns                         1
##
## Model Test User Model:
##
##   Test statistic                                99.587
##   Degrees of freedom                                 5
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                              1190.537
##   Degrees of freedom                                10
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.920
##   Tucker-Lewis Index (TLI)                       0.840
##
##   Robust Comparative Fit Index (CFI)             0.920
##   Robust Tucker-Lewis Index (TLI)                0.840
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1580.254
##   Loglikelihood unrestricted model (H1)      -1530.461
##
##   Akaike (AIC)                                3190.508
##   Bayesian (BIC)                              3252.080
##   Sample-size adjusted Bayesian (SABIC)       3204.476
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.205
##   90 Percent confidence interval - lower         0.171
##   90 Percent confidence interval - upper         0.242
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    1.000
##
##   Robust RMSEA                                   0.205
##   90 Percent confidence interval - lower         0.171
##   90 Percent confidence interval - upper         0.242
##   P-value H_0: Robust RMSEA <= 0.050             0.000
##   P-value H_0: Robust RMSEA >= 0.080             1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.048
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.922
##   90 Percent confidence interval - lower         0.895
##   90 Percent confidence interval - upper         0.945
##
##   Robust GFI                                     0.922
##   90 Percent confidence interval - lower         0.895
##   90 Percent confidence interval - upper         0.945
##
## Parameter Estimates:
##
##   Standard errors                             Standard
```

```
##   Information                                 Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Evaluation =~
##     hon               1.000                               0.455    0.788
##     sin               1.108    0.061   18.065    0.000    0.504    0.812
##     eth               1.089    0.058   18.896    0.000    0.495    0.830
##     like              1.098    0.069   16.033    0.000    0.499    0.768
##     app               0.958    0.074   12.942    0.000    0.436    0.634
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               1.885    0.027   69.161    0.000    1.885    3.268
##    .sin               2.099    0.029   71.616    0.000    2.099    3.384
##    .eth               1.877    0.028   66.587    0.000    1.877    3.146
##    .like              2.273    0.031   73.986    0.000    2.273    3.496
##    .app               2.661    0.032   81.946    0.000    2.661    3.872
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.126    0.011   11.453    0.000    0.126    0.379
##    .sin               0.131    0.012   11.070    0.000    0.131    0.341
##    .eth               0.111    0.011   10.381    0.000    0.111    0.311
##    .like              0.173    0.015   11.699    0.000    0.173    0.410
##    .app               0.283    0.021   13.374    0.000    0.283    0.599
##     Evaluation        0.207    0.022    9.503    0.000    1.000    1.000
```

```
cat("\n================ INVESTIGATOR: TWO FACTORS ================\n")
```

```
##
## ================ INVESTIGATOR: TWO FACTORS ================
```

```
summary(
  investigator_two,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 28 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        16
##
##                                                   Used       Total
##   Number of observations                           448         455
##   Number of missing patterns                         1
##
## Model Test User Model:
##
##   Test statistic                                26.909
##   Degrees of freedom                                 4
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                              1190.537
##   Degrees of freedom                                10
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.981
##   Tucker-Lewis Index (TLI)                       0.951
##
##   Robust Comparative Fit Index (CFI)             0.981
##   Robust Tucker-Lewis Index (TLI)                0.951
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1543.915
##   Loglikelihood unrestricted model (H1)      -1530.461
##
##   Akaike (AIC)                                3119.830
##   Bayesian (BIC)                              3185.507
##   Sample-size adjusted Bayesian (SABIC)       3134.729
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.113
##   90 Percent confidence interval - lower         0.075
##   90 Percent confidence interval - upper         0.155
##   P-value H_0: RMSEA <= 0.050                    0.004
##   P-value H_0: RMSEA >= 0.080                    0.925
##
##   Robust RMSEA                                   0.113
##   90 Percent confidence interval - lower         0.075
##   90 Percent confidence interval - upper         0.155
##   P-value H_0: Robust RMSEA <= 0.050             0.004
##   P-value H_0: Robust RMSEA >= 0.080             0.925
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.022
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.980
##   90 Percent confidence interval - lower         0.963
##   90 Percent confidence interval - upper         0.991
##
##   Robust GFI                                     0.980
##   90 Percent confidence interval - lower         0.963
##   90 Percent confidence interval - upper         0.991
##
## Parameter Estimates:
##
##   Standard errors                             Standard
```

```
##   Information                                 Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.468    0.812
##     sin               1.072    0.058   18.374    0.000    0.502    0.810
##     eth               1.079    0.055   19.557    0.000    0.505    0.847
##   Likeability =~
##     like              1.000                               0.594    0.913
##     app               0.829    0.057   14.542    0.000    0.492    0.716
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.221    0.020   11.172    0.000    0.793    0.793
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               1.885    0.027   69.161    0.000    1.885    3.268
##    .sin               2.099    0.029   71.616    0.000    2.099    3.384
##    .eth               1.877    0.028   66.587    0.000    1.877    3.146
##    .like              2.273    0.031   73.986    0.000    2.273    3.496
##    .app               2.661    0.032   81.946    0.000    2.661    3.872
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.113    0.010   10.896    0.000    0.113    0.340
##    .sin               0.132    0.012   10.620    0.000    0.132    0.344
##    .eth               0.100    0.011    9.401    0.000    0.100    0.282
##    .like              0.070    0.019    3.786    0.000    0.070    0.166
##    .app               0.230    0.020   11.670    0.000    0.230    0.487
##     Integrity         0.219    0.022    9.938    0.000    1.000    1.000
##     Likeability       0.353    0.033   10.641    0.000    1.000    1.000
```

```
cat("\n================ INSPECTOR: ONE FACTOR ================\n")
```

```
##
## ================ INSPECTOR: ONE FACTOR ================
```

```
summary(
  inspector_one,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 46 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        15
##
##                                                   Used       Total
##   Number of observations                           447         455
##   Number of missing patterns                         1
##
## Model Test User Model:
##
##   Test statistic                               130.573
##   Degrees of freedom                                 5
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                              1273.192
##   Degrees of freedom                                10
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.901
##   Tucker-Lewis Index (TLI)                       0.801
##
##   Robust Comparative Fit Index (CFI)             0.901
##   Robust Tucker-Lewis Index (TLI)                0.801
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1884.593
##   Loglikelihood unrestricted model (H1)      -1819.306
##
##   Akaike (AIC)                                3799.185
##   Bayesian (BIC)                              3860.724
##   Sample-size adjusted Bayesian (SABIC)       3813.120
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.237
##   90 Percent confidence interval - lower         0.203
##   90 Percent confidence interval - upper         0.273
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    1.000
##
##   Robust RMSEA                                   0.237
##   90 Percent confidence interval - lower         0.203
##   90 Percent confidence interval - upper         0.273
##   P-value H_0: Robust RMSEA <= 0.050             0.000
##   P-value H_0: Robust RMSEA >= 0.080             1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.053
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.899
##   90 Percent confidence interval - lower         0.870
##   90 Percent confidence interval - upper         0.924
##
##   Robust GFI                                     0.899
##   90 Percent confidence interval - lower         0.870
##   90 Percent confidence interval - upper         0.924
##
## Parameter Estimates:
##
##   Standard errors                             Standard
```

```
##   Information                                 Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Evaluation =~
##     hon               1.000                               0.454    0.728
##     sin               1.374    0.080   17.118    0.000    0.623    0.838
##     eth               1.323    0.079   16.669    0.000    0.600    0.810
##     like              1.354    0.089   15.149    0.000    0.614    0.801
##     app               1.209    0.090   13.506    0.000    0.549    0.715
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               4.482    0.029  152.144    0.000    4.482    7.196
##    .sin               4.186    0.035  118.985    0.000    4.186    5.628
##    .eth               4.250    0.035  121.285    0.000    4.250    5.737
##    .like              3.838    0.036  105.867    0.000    3.838    5.007
##    .app               3.769    0.036  103.881    0.000    3.769    4.913
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.182    0.015   12.432    0.000    0.182    0.469
##    .sin               0.165    0.016   10.257    0.000    0.165    0.298
##    .eth               0.189    0.017   11.061    0.000    0.189    0.344
##    .like              0.210    0.020   10.641    0.000    0.210    0.358
##    .app               0.287    0.024   12.224    0.000    0.287    0.488
##     Evaluation        0.206    0.024    8.467    0.000    1.000    1.000
```

```
cat("\n================ INSPECTOR: TWO FACTORS ================\n")
```

```
##
## ================ INSPECTOR: TWO FACTORS ================
```

```
summary(
  inspector_two,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 47 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        16
##
##                                                   Used       Total
##   Number of observations                           447         455
##   Number of missing patterns                         1
##
## Model Test User Model:
##
##   Test statistic                                15.058
##   Degrees of freedom                                 4
##   P-value (Chi-square)                           0.005
##
## Model Test Baseline Model:
##
##   Test statistic                              1273.192
##   Degrees of freedom                                10
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.991
##   Tucker-Lewis Index (TLI)                       0.978
##
##   Robust Comparative Fit Index (CFI)             0.991
##   Robust Tucker-Lewis Index (TLI)                0.978
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1826.835
##   Loglikelihood unrestricted model (H1)      -1819.306
##
##   Akaike (AIC)                                3685.670
##   Bayesian (BIC)                              3751.311
##   Sample-size adjusted Bayesian (SABIC)       3700.534
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.079
##   90 Percent confidence interval - lower         0.039
##   90 Percent confidence interval - upper         0.123
##   P-value H_0: RMSEA <= 0.050                    0.107
##   P-value H_0: RMSEA >= 0.080                    0.532
##
##   Robust RMSEA                                   0.079
##   90 Percent confidence interval - lower         0.039
##   90 Percent confidence interval - upper         0.123
##   P-value H_0: Robust RMSEA <= 0.050             0.107
##   P-value H_0: Robust RMSEA >= 0.080             0.532
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.019
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.990
##   90 Percent confidence interval - lower         0.976
##   90 Percent confidence interval - upper         0.998
##
##   Robust GFI                                     0.990
##   90 Percent confidence interval - lower         0.976
##   90 Percent confidence interval - upper         0.998
##
## Parameter Estimates:
##
##   Standard errors                             Standard
```

```
##   Information                                 Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.475    0.763
##     sin               1.338    0.075   17.863    0.000    0.636    0.855
##     eth               1.295    0.073   17.729    0.000    0.615    0.831
##   Likeability =~
##     like              1.000                               0.714    0.931
##     app               0.855    0.047   18.102    0.000    0.610    0.795
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.268    0.025   10.895    0.000    0.789    0.789
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               4.482    0.029  152.144    0.000    4.482    7.196
##    .sin               4.186    0.035  118.985    0.000    4.186    5.628
##    .eth               4.250    0.035  121.284    0.000    4.250    5.737
##    .like              3.838    0.036  105.867    0.000    3.838    5.007
##    .app               3.769    0.036  103.881    0.000    3.769    4.913
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.162    0.013   12.036    0.000    0.162    0.417
##    .sin               0.149    0.017    8.958    0.000    0.149    0.269
##    .eth               0.170    0.017    9.900    0.000    0.170    0.310
##    .like              0.078    0.021    3.686    0.000    0.078    0.132
##    .app               0.216    0.021   10.397    0.000    0.216    0.367
##     Integrity         0.226    0.025    9.081    0.000    1.000    1.000
##     Likeability       0.510    0.044   11.586    0.000    1.000    1.000
```

```
# 5. Compare one-factor vs. two-factor models


cat("\n================ MODEL COMPARISONS ================\n")
```

```
##
## ================ MODEL COMPARISONS ================
```

```
cat("\nInvestigator:\n")
```

```
##
## Investigator:
```

```
anova(
  investigator_one,
  investigator_two
)
```

```
##
## Chi-Squared Difference Test
##
##                  Df    AIC    BIC  Chisq Chisq diff Df diff Pr(>Chisq)
## investigator_two  4 3119.8 3185.5 26.909
## investigator_one  5 3190.5 3252.1 99.587     72.678       1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\nInspector:\n")
```

```
##
## Inspector:
```

```
anova(
  inspector_one,
  inspector_two
)
```

```
##
## Chi-Squared Difference Test
##
##               Df    AIC    BIC   Chisq Chisq diff Df diff Pr(>Chisq)
## inspector_two  4 3685.7 3751.3  15.058
## inspector_one  5 3799.2 3860.7 130.573     115.52       1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ============================================================
# 6. Create easy-to-read fit table
# ============================================================

fit_table <- rbind(
  Investigator_1Factor = fitMeasures(
    investigator_one,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  ),

  Investigator_2Factor = fitMeasures(
    investigator_two,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  ),

  Inspector_1Factor = fitMeasures(
    inspector_one,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  ),

  Inspector_2Factor = fitMeasures(
    inspector_two,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  )
)

cat("\n================ FIT TABLE ================\n")
```

```
##
## ================ FIT TABLE ================
```

```
print(round(fit_table, 3))
```

```
##                       chisq df   cfi   tli rmsea  srmr
## Investigator_1Factor  99.587  5 0.920 0.840 0.205 0.048
## Investigator_2Factor  26.909  4 0.981 0.951 0.113 0.022
## Inspector_1Factor    130.573  5 0.901 0.801 0.237 0.053
## Inspector_2Factor     15.058  4 0.991 0.978 0.079 0.019
```

```
# ============================================================
# 7. Measurement invariance across researcher role
# ============================================================
# ONLY run/interpret this section if the two-factor model
# provides a reasonable fit.
#
# Configural = same factor structure
# Metric     = equal factor loadings
# Scalar     = equal loadings + intercepts


cat("\n================ CONFIGURAL INVARIANCE ================\n")
```

```
##
## ================ CONFIGURAL INVARIANCE ================
```

```
configural <- cfa(
  two_factor,
  data = cfa_dat,
  group = "role",
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 113 215 367 557 765 839 861.
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 114 216 344 368 558 766 840 862.
```

```
summary(
  configural,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 59 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        32
##
##   Number of observations per group:               Used       Total
##     investigator                                   448         455
##     inspector                                      447         455
##   Number of missing patterns per group:
##     investigator                                     1
##     inspector                                        1
##
## Model Test User Model:
##
##   Test statistic                                41.967
##   Degrees of freedom                                 8
##   P-value (Chi-square)                           0.000
##   Test statistic for each group:
##     investigator                                26.909
##     inspector                                   15.058
##
## Model Test Baseline Model:
##
##   Test statistic                              2463.729
##   Degrees of freedom                                20
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.986
##   Tucker-Lewis Index (TLI)                       0.965
##
##   Robust Comparative Fit Index (CFI)             0.986
##   Robust Tucker-Lewis Index (TLI)                0.965
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3370.750
##   Loglikelihood unrestricted model (H1)      -3349.767
##
##   Akaike (AIC)                                6805.501
##   Bayesian (BIC)                              6958.999
##   Sample-size adjusted Bayesian (SABIC)       6857.373
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.097
##   90 Percent confidence interval - lower         0.070
##   90 Percent confidence interval - upper         0.127
##   P-value H_0: RMSEA <= 0.050                    0.003
##   P-value H_0: RMSEA >= 0.080                    0.856
##
##   Robust RMSEA                                   0.097
##   90 Percent confidence interval - lower         0.070
##   90 Percent confidence interval - upper         0.127
##   P-value H_0: Robust RMSEA <= 0.050             0.003
##   P-value H_0: Robust RMSEA >= 0.080             0.856
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.021
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.985
##   90 Percent confidence interval - lower         0.975
##   90 Percent confidence interval - upper         0.992
##
##   Robust GFI                                     0.985
```

```
##   90 Percent confidence interval - lower         0.975
##   90 Percent confidence interval - upper         0.992
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Observed
##   Observed information based on                Hessian
##
##
## Group 1 [investigator]:
##
## Latent Variables:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.468    0.812
##     sin               1.072    0.058   18.374    0.000    0.502    0.810
##     eth               1.079    0.055   19.557    0.000    0.505    0.847
##   Likeability =~
##     like              1.000                               0.594    0.913
##     app               0.829    0.057   14.542    0.000    0.492    0.716
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.221    0.020   11.172    0.000    0.793    0.793
##
## Intercepts:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               1.885    0.027   69.161    0.000    1.885    3.268
##    .sin               2.099    0.029   71.616    0.000    2.099    3.384
##    .eth               1.877    0.028   66.587    0.000    1.877    3.146
##    .like              2.273    0.031   73.986    0.000    2.273    3.496
##    .app               2.661    0.032   81.946    0.000    2.661    3.872
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.113    0.010   10.896    0.000    0.113    0.340
##    .sin               0.132    0.012   10.620    0.000    0.132    0.344
##    .eth               0.100    0.011    9.401    0.000    0.100    0.282
##    .like              0.070    0.019    3.786    0.000    0.070    0.166
##    .app               0.230    0.020   11.670    0.000    0.230    0.487
##     Integrity         0.219    0.022    9.938    0.000    1.000    1.000
##     Likeability       0.353    0.033   10.641    0.000    1.000    1.000
##
##
## Group 2 [inspector]:
##
## Latent Variables:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.475    0.763
##     sin               1.338    0.075   17.863    0.000    0.636    0.855
##     eth               1.295    0.073   17.729    0.000    0.615    0.831
##   Likeability =~
##     like              1.000                               0.714    0.931
##     app               0.855    0.047   18.102    0.000    0.610    0.795
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.268    0.025   10.895    0.000    0.789    0.789
##
## Intercepts:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               4.482    0.029  152.144    0.000    4.482    7.196
##    .sin               4.186    0.035  118.985    0.000    4.186    5.628
##    .eth               4.250    0.035  121.285    0.000    4.250    5.737
##    .like              3.838    0.036  105.867    0.000    3.838    5.007
##    .app               3.769    0.036  103.881    0.000    3.769    4.913
##
```

```
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.162    0.013   12.036    0.000    0.162    0.417
##    .sin               0.149    0.017    8.958    0.000    0.149    0.269
##    .eth               0.170    0.017    9.900    0.000    0.170    0.310
##    .like              0.078    0.021    3.686    0.000    0.078    0.132
##    .app               0.216    0.021   10.397    0.000    0.216    0.367
##     Integrity         0.226    0.025    9.081    0.000    1.000    1.000
##     Likeability       0.510    0.044   11.586    0.000    1.000    1.000
```

```
cat("\n================ METRIC INVARIANCE ================\n")
```

```
##
## ================ METRIC INVARIANCE ================
```

```
metric <- cfa(
  two_factor,
  data = cfa_dat,
  group = "role",
  group.equal = "loadings",
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 113 215 367 557 765 839 861.
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 114 216 344 368 558 766 840 862.
```

```
summary(
  metric,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 51 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        32
##   Number of equality constraints                     3
##
##   Number of observations per group:               Used       Total
##     investigator                                   448         455
##     inspector                                      447         455
##   Number of missing patterns per group:
##     investigator                                     1
##     inspector                                        1
##
## Model Test User Model:
##
##   Test statistic                                51.278
##   Degrees of freedom                                11
##   P-value (Chi-square)                           0.000
##   Test statistic for each group:
##     investigator                                31.790
##     inspector                                   19.489
##
## Model Test Baseline Model:
##
##   Test statistic                              2463.729
##   Degrees of freedom                                20
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.984
##   Tucker-Lewis Index (TLI)                       0.970
##
##   Robust Comparative Fit Index (CFI)             0.984
##   Robust Tucker-Lewis Index (TLI)                0.970
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3375.406
##   Loglikelihood unrestricted model (H1)      -3349.767
##
##   Akaike (AIC)                                6808.812
##   Bayesian (BIC)                              6947.920
##   Sample-size adjusted Bayesian (SABIC)       6855.821
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.090
##   90 Percent confidence interval - lower         0.066
##   90 Percent confidence interval - upper         0.116
##   P-value H_0: RMSEA <= 0.050                    0.004
##   P-value H_0: RMSEA >= 0.080                    0.776
##
##   Robust RMSEA                                   0.090
##   90 Percent confidence interval - lower         0.066
##   90 Percent confidence interval - upper         0.116
##   P-value H_0: Robust RMSEA <= 0.050             0.004
##   P-value H_0: Robust RMSEA >= 0.080             0.776
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.035
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.982
##   90 Percent confidence interval - lower         0.971
##   90 Percent confidence interval - upper         0.990
##
```

```
##   Robust GFI                                        0.982
##   90 Percent confidence interval - lower            0.971
##   90 Percent confidence interval - upper            0.990
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Observed
##   Observed information based on                Hessian
##
##
## Group 1 [investigator]:
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.434    0.781
##     sin     (.p2.)    1.201    0.047   25.407    0.000    0.521    0.824
##     eth     (.p3.)    1.181    0.045   25.974    0.000    0.513    0.851
##   Likeability =~
##     like              1.000                               0.591    0.910
##     app     (.p5.)    0.844    0.036   23.199    0.000    0.498    0.722
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.205    0.018   11.421    0.000    0.800    0.800
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               1.885    0.026   71.772    0.000    1.885    3.391
##    .sin               2.099    0.030   70.171    0.000    2.099    3.315
##    .eth               1.877    0.028   65.912    0.000    1.877    3.114
##    .like              2.273    0.031   74.099    0.000    2.273    3.501
##    .app               2.661    0.033   81.599    0.000    2.661    3.855
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.120    0.010   11.577    0.000    0.120    0.390
##    .sin               0.129    0.012   10.499    0.000    0.129    0.321
##    .eth               0.100    0.011    9.490    0.000    0.100    0.276
##    .like              0.073    0.016    4.513    0.000    0.073    0.173
##    .app               0.228    0.019   12.197    0.000    0.228    0.479
##     Integrity         0.188    0.018   10.526    0.000    1.000    1.000
##     Likeability       0.349    0.031   11.380    0.000    1.000    1.000
##
##
## Group 2 [inspector]:
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.512    0.791
##     sin     (.p2.)    1.201    0.047   25.407    0.000    0.615    0.842
##     eth     (.p3.)    1.181    0.045   25.974    0.000    0.605    0.826
##   Likeability =~
##     like              1.000                               0.717    0.934
##     app     (.p5.)    0.844    0.036   23.199    0.000    0.605    0.791
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.288    0.025   11.509    0.000    0.784    0.784
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               4.482    0.031  146.415    0.000    4.482    6.925
##    .sin               4.186    0.035  121.125    0.000    4.186    5.729
##    .eth               4.250    0.035  122.686    0.000    4.250    5.803
##    .like              3.838    0.036  105.749    0.000    3.838    5.002
##    .app               3.769    0.036  104.223    0.000    3.769    4.930
```

```
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.157    0.013   11.746    0.000    0.157    0.374
##    .sin               0.156    0.016    9.825    0.000    0.156    0.291
##    .eth               0.170    0.016   10.468    0.000    0.170    0.318
##    .like              0.075    0.020    3.783    0.000    0.075    0.127
##    .app               0.218    0.020   10.902    0.000    0.218    0.374
##     Integrity         0.262    0.024   10.918    0.000    1.000    1.000
##     Likeability       0.514    0.042   12.125    0.000    1.000    1.000
```

```
cat("\n================ SCALAR INVARIANCE ================\n")
```

```
##
## ================ SCALAR INVARIANCE ================
```

```
scalar <- cfa(
  two_factor,
  data = cfa_dat,
  group = "role",
  group.equal = c("loadings", "intercepts"),
  missing = "fiml"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 113 215 367 557 765 839 861.
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 114 216 344 368 558 766 840 862.
```

```
summary(
  scalar,
  fit.measures = TRUE,
  standardized = TRUE
)
```

```
## lavaan 0.7-2 ended normally after 65 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        34
##   Number of equality constraints                     8
##
##   Number of observations per group:               Used       Total
##     investigator                                   448         455
##     inspector                                      447         455
##   Number of missing patterns per group:
##     investigator                                     1
##     inspector                                        1
##
## Model Test User Model:
##
##   Test statistic                               157.985
##   Degrees of freedom                                14
##   P-value (Chi-square)                           0.000
##   Test statistic for each group:
##     investigator                                 54.966
##     inspector                                   103.020
##
## Model Test Baseline Model:
##
##   Test statistic                              2463.729
##   Degrees of freedom                                20
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.941
##   Tucker-Lewis Index (TLI)                       0.916
##
##   Robust Comparative Fit Index (CFI)             0.941
##   Robust Tucker-Lewis Index (TLI)                0.916
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3428.760
##   Loglikelihood unrestricted model (H1)      -3349.767
##
##   Akaike (AIC)                                6909.519
##   Bayesian (BIC)                              7034.236
##   Sample-size adjusted Bayesian (SABIC)       6951.665
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.152
##   90 Percent confidence interval - lower         0.131
##   90 Percent confidence interval - upper         0.173
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    1.000
##
##   Robust RMSEA                                   0.152
##   90 Percent confidence interval - lower         0.131
##   90 Percent confidence interval - upper         0.173
##   P-value H_0: Robust RMSEA <= 0.050             0.000
##   P-value H_0: Robust RMSEA >= 0.080             1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.097
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.940
##   90 Percent confidence interval - lower         0.922
##   90 Percent confidence interval - upper         0.954
##
```

```
##   Robust GFI                                  0.940
##   90 Percent confidence interval - lower      0.922
##   90 Percent confidence interval - upper      0.954
##
## Parameter Estimates:
##
##   Standard errors                           Standard
##   Information                               Observed
##   Observed information based on              Hessian
##
##
## Group 1 [investigator]:
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.521    0.848
##     sin     (.p2.)    0.848    0.013   66.751    0.000    0.442    0.755
##     eth     (.p3.)    0.946    0.012   75.820    0.000    0.493    0.841
##   Likeability =~
##     like              1.000                               0.613    0.938
##     app     (.p5.)    0.744    0.019   39.134    0.000    0.456    0.679
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.245    0.021   11.723    0.000    0.768    0.768
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon     (.14.)    1.910    0.029   65.687    0.000    1.910    3.108
##    .sin     (.15.)    2.072    0.027   75.522    0.000    2.072    3.537
##    .eth     (.16.)    1.868    0.028   67.843    0.000    1.868    3.190
##    .like    (.17.)    2.277    0.031   73.638    0.000    2.277    3.483
##    .app     (.18.)    2.635    0.031   86.143    0.000    2.635    3.921
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.106    0.011    9.898    0.000    0.106    0.281
##    .sin               0.148    0.012   12.419    0.000    0.148    0.431
##    .eth               0.100    0.010   10.338    0.000    0.100    0.292
##    .like              0.052    0.017    3.024    0.002    0.052    0.121
##    .app               0.244    0.019   13.025    0.000    0.244    0.539
##     Integrity         0.271    0.021   12.657    0.000    1.000    1.000
##     Likeability       0.376    0.031   12.137    0.000    1.000    1.000
##
##
## Group 2 [inspector]:
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity =~
##     hon               1.000                               0.598    0.838
##     sin     (.p2.)    0.848    0.013   66.751    0.000    0.508    0.752
##     eth     (.p3.)    0.946    0.012   75.820    0.000    0.566    0.803
##   Likeability =~
##     like              1.000                               0.744    0.965
##     app     (.p5.)    0.744    0.019   39.134    0.000    0.554    0.746
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Integrity ~~
##     Likeability       0.336    0.029   11.561    0.000    0.755    0.755
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon     (.14.)    1.910    0.029   65.687    0.000    1.910    2.675
##    .sin     (.15.)    2.072    0.027   75.522    0.000    2.072    3.069
##    .eth     (.16.)    1.868    0.028   67.843    0.000    1.868    2.650
##    .like    (.17.)    2.277    0.031   73.638    0.000    2.277    2.954
##    .app     (.18.)    2.635    0.031   86.143    0.000    2.635    3.554
```

```
##     Intgrty           2.535    0.044   57.030    0.000    4.238    4.238
##     Likblty           1.558    0.048   32.550    0.000    2.095    2.095
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hon               0.152    0.016    9.625    0.000    0.152    0.298
##    .sin               0.198    0.016   12.002    0.000    0.198    0.434
##    .eth               0.177    0.016   10.989    0.000    0.177    0.356
##    .like              0.041    0.020    2.003    0.045    0.041    0.069
##    .app               0.244    0.020   12.270    0.000    0.244    0.443
##     Integrity         0.358    0.029   12.199    0.000    1.000    1.000
##     Likeability       0.553    0.042   13.121    0.000    1.000    1.000
```

```
# 8. Compare invariance models

cat("\n================ INVARIANCE MODEL COMPARISON ================\n")
```

```
##
## ================ INVARIANCE MODEL COMPARISON ================
```

```
anova(
  configural,
  metric,
  scalar
)
```

```
##
## Chi-Squared Difference Test
##
##            Df    AIC    BIC   Chisq Chisq diff Df diff Pr(>Chisq)
## configural  8 6805.5 6959.0  41.967
## metric     11 6808.8 6947.9  51.279      9.312       3    0.02542 *
## scalar     14 6909.5 7034.2 157.986    106.707       3    < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 9. Invariance fit table

invariance_fit_table <- rbind(
  Configural = fitMeasures(
    configural,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  ),

  Metric = fitMeasures(
    metric,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  ),

  Scalar = fitMeasures(
    scalar,
    c("chisq", "df", "cfi", "tli", "rmsea", "srmr")
  )
)

cat("\n================ INVARIANCE FIT TABLE ================\n")
```

```
##
## ================ INVARIANCE FIT TABLE ================
```

```
print(round(invariance_fit_table, 3))
```

```
##              chisq df   cfi   tli rmsea  srmr
## Configural  41.967  8 0.986 0.965 0.097 0.021
## Metric      51.278 11 0.984 0.970 0.090 0.035
## Scalar     157.985 14 0.941 0.916 0.152 0.097
```